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FACULTY OF ENGINEERING AND TECHNOLOGY

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DESIGN AND IMPLEMENTATION OF A LOW-COST OBJECT DETECTION SYSTEM FOR WASTE SORTING AND RECYCLING (WASTEDETECT)

A dissertation submitted to the Department of Computer Engineering, Faculty of Engineering and Technology, University of Buea, in Partial Fulfilment of the Requirements for the Award of Bachelor of Engineering (B.Eng.) Degree in Computer Engineering.

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Department of Computer Engineering

Faculty of Engineering and Technology

University of Buea

I. CERTIFICATE OF ORIGINALITY

We the undersigned, hereby certify that this dissertation entitled "**Design and Implementation of a Low-Cost Object Detection System for Waste Sorting and Recycling (WasteDetect)**" presented by ATANKEU TCHAKOUTE Ange Natanahel, matriculation number **FE22A158** has been carried out by him in the Department of Computer Engineering, Faculty of Engineering and Technology, University of Buea under the supervision of **Dr. SOP DEFFO Lionel Landry.**

This dissertation is authentic and represents the fruits of his own research and efforts.

Date______________________

Student **Supervisor**

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Head of Department

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II. DEDICATION

I dedicate this project to my family; the TCHAKOUTE's family.

III. ACKNOWLEDGEMENT

I express my deepest gratitude to Almighty God and to my family for their unwavering love, encouragement, and sacrifices throughout my undergraduate studies. I am especially indebted to my father, Mr. TCHAKOUTE Pierre, for his continuous support, and to my mother, Mme TCHAKOUTE Henriette, for her steadfast prayers and encouragement. I also sincerely appreciate my elder brothers, Mr. TCHANOU Rodolph and Mr. KODJO Leroy, for their invaluable advice and support throughout my academic journey. My heartfelt thanks go to my Elder Sister, Miss CHAMDA Manuela for her remarkable financial, emotional, and moral support, and to Mr. AGBOR Derrick OBI for his endless guidance, encouragement, and unwavering support, all of which greatly contributed to the successful completion of this project.

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IV. ABSTRACT

Waste management in Cameroon remains largely manual, with no institutional sorting mechanism in place at organisations such as HYSACAM, while existing AI-based waste sorting systems documented in the literature depend on expensive infrastructure; GPUs, robotic arms, and conveyor systems impractical for resource-constrained environments. This study addresses that gap by designing and implementing WasteDetect, a low-cost, human-assisted object detection system for waste sorting and recycling, built using lightweight deep learning techniques suitable for resource-constrained settings such as Buea, Cameroon. In the research phase, a structured field investigation at HYSACAM, Buea, confirmed the absence of any systematic sorting mechanism and identified plastic, metal, and organic waste as the categories of greatest operational interest, with paper and glass as desirable additions. These findings, together with a brainstorming phase weighing implementation, technical, deployment, and maintenance costs, established a five-class waste category and a low-cost design direction. A YOLOv8n model was fine-tuned via transfer learning on a re-split, class-remapped Kaggle dataset of 10,464 images, with the training split expanded to 43,869 images through five augmentation pipelines simulating real world lighting and camera variability. Post-training INT8 quantization via OpenVINO reduced the model's file size from 17.52 MB to 3.00 MB (an 82.9% reduction) while limiting the drop in accuracy (mAP@50) to just 1.4%. On a held-out test split, the final INT8 model achieved an accuracy (mAP@50) of 0.598, comparing favourably with published YOLOv8-based waste detection systems. The optimised model was integrated into a Flask backend serving a JWT-authenticated REST API, paired with a React PWA that captures waste images via WebRTC and returns a predicted category, confidence score, and bin placement instruction, with offline-first support queuing images for auto-submission once connectivity is restored. Benchmarked against published systems, WasteDetect achieves classification performance broadly comparable to systems built on heavier models and more expensive infrastructure, while remaining substantially cheaper to build, deploy, and maintain. The study concludes that this low-cost, software-centred approach is a practical contribution to waste management in developing country contexts, and identifies completion of admin dashboard functionality and locally collected training data as priority future work.

Keywords: Waste Sorting, Object Detection, YOLOv8n, Transfer Learning, Image Augmentation, INT8 Quantization, HYSACAM.

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Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
CNN	Convolutional Neural Network
COCO	Common Objects in Context (dataset)
CPU	Central Processing Unit
CUDA	Compute Unified Device Architecture
DB	Database
Dev Database	Development Database
ERD	Entity Relationship Diagram
FP32	Floating Point 32-bit (single-precision floating point format)
Flask-JWT-Extended	Flask JSON Web Token Extended (Flask authentication library)
GPU	Graphics Processing Unit
HTTPS	Hypertext Transfer Protocol Secure
HYSACAM	Hygiène et Salubrité du Cameroun (Cameroon Hygiene and Sanitation Company)
ID	Identifier
INT8	8-bit Integer (quantized numerical precision format)
JSON	JavaScript Object Notation
JWT	JSON Web Token

Abbreviation	Meaning
mAP@50	Mean Average Precision at 50% Intersection over Union threshold
MobileNet	Mobile Network (lightweight convolutional neural network architecture)
MVP	Minimum Viable Product
NVIDIA	NVIDIA Corporation (graphics and AI hardware manufacturer)
OpenCV	Open Source Computer Vision Library
OpenVINO	Open Visual Inference and Neural Network Optimization (toolkit)
OpenVINO IR	OpenVINO Intermediate Representation
Prod Database	Production Database
PWA	Progressive Web Application
R-CNN	Region-based Convolutional Neural Network
RAM	Random Access Memory
RBAC	Role-Based Access Control
REST API	Representational State Transfer Application Programming Interface
SDLC	Software Development Life Cycle
ShuffleNet	Shuffle Network (lightweight convolutional neural network architecture)
SQL	Structured Query Language
SQLite	Structured Query Language Lite (lightweight embedded database engine)

Abbreviation	Meaning
SqueezeNet	Squeeze Network (lightweight convolutional neural network architecture)
SSD	Single Shot Detector
Tesla T4	NVIDIA Tesla T4 (data-centre graphics processing unit model)
UI	User Interface
URL	Uniform Resource Locator
WaRP	Waste Recycling Plant (dataset)
WebRTC	Web Real time Communication
YOLO	You Only Look Once (object detection algorithm)
YOLOv8	You Only Look Once, version 8
YOLOv8n	You Only Look Once, version 8, nano variant
YOLOv8s	You Only Look Once, version 8, small variant

CHAPTER 1. GENERAL INTRODUCTION

1.1 BACKGROUND AND CONTEXT OF THE STUDY

Waste generation has become a major global challenge, driven by rapid urbanization, industrialization, and population growth. According to the World Bank, global waste generation is projected to increase from 2.01 billion tonnes in 2016 to 3.40 billion tonnes by 2050, placing significant pressure on existing waste management systems [1]. In response, many developed countries have adopted artificial intelligence (AI)-driven automated sorting systems, integrating computer vision, robotics, and deep learning to improve efficiency and accuracy in waste classification. Recent studies indicate that such systems can achieve classification accuracies exceeding 80–90%, significantly outperforming traditional manual sorting methods in terms of speed, consistency, and safety [2].

In contrast, waste management practices in developing countries, particularly in Cameroon, remain largely manual or semi-structured. Organizations such as HYSACAM are responsible for waste collection and disposal in major cities like Douala and Yaoundé, yet systematic sorting is limited. Reports indicate that Cameroon generates around 16,000 tons of municipal solid waste daily [3], with significant amounts either openly dumped, burned, or poorly managed due to inadequate infrastructure and weak enforcement of environmental policies [4]. In Cameroon specifically, rapid urban growth has led to increasing waste generation, with only approximately 486 tons of waste effectively recycled daily [5], largely due to inefficient collection and sorting practices [6].

One of the key challenges contributing to this situation is the lack of waste separation at the source, including households, institutions, and public spaces, which complicates downstream sorting efforts. Additionally, waste collection systems often mix different waste types during transportation, further reducing the potential for recycling. Research highlights that financial constraints, lack of technological adoption, and limited public awareness are major barriers to effective waste management systems in developing countries [4].

Despite various initiatives aimed at improving waste management practices in Cameroon, progress remains limited. The absence of localized, cost effective technological solutions, particularly those leveraging AI for real time waste classification, continues to hinder advancements in recycling efficiency. Most existing AI-based waste sorting systems are designed for high-resource environments and rely on expensive infrastructure such as conveyor belts, robotic arms, and high-performance computing systems, making them unsuitable for deployment in resource-constrained settings [7].

This gap underscores the need for low-cost, context-aware intelligent systems that can operate effectively within the constraints of developing regions. In this regard, a human-assisted object detection system for waste sorting presents a potential practical and scalable solution. By leveraging lightweight deep learning models and accessible technologies such as mobile devices and web-based applications, such systems can support users in identifying and sorting waste more efficiently, thereby improving recycling rates and contributing to environmental sustainability.

1.2 PROBLEM STATEMENT

Effective waste sorting is a critical step in achieving sustainable waste management and improving recycling outcomes. However, in many developing countries, including Cameroon, waste sorting practices remain inefficient and inconsistently applied, largely due to limited automation and reliance on manual processes. Studies indicate that inefficient waste sorting significantly reduces recycling efficiency and leads to poor environmental outcomes, as mixed waste streams are difficult to process and recover [4]. In such contexts, the absence of structured and standardized sorting mechanisms results in a large proportion of recyclable materials being lost.

Although recent advancements in artificial intelligence (AI) have introduced automated waste sorting systems capable of high accuracy and real time performance, these systems are often expensive and computationally demanding, making them unsuitable for deployment in resource-constrained environments. Many AI-based solutions rely on high-performance hardware, including GPUs, robotic arms, and conveyor-based infrastructures, which significantly increase implementation and maintenance costs [7]. Consequently, such technologies remain largely inaccessible to waste management systems in developing regions.

Another major challenge is the limited availability of diverse and well-annotated waste datasets suitable for developing robust object detection systems. Existing public datasets often contain limitations such as class imbalance and restricted environmental variability, which may affect model performance in real world scenarios [8]. Furthermore, differences in waste composition and environmental conditions across regions can affect the adaptability of AI-based systems when deployed in different contexts [2].

As a result of these challenges, recycling efficiency remains low, and scalable, cost effective solutions for waste sorting are still limited in many developing countries, including Cameroon. Mixed waste streams, inadequate classification mechanisms, and limited technological support continue to hinder effective recycling processes and environmental sustainability efforts [4]. Addressing these issues requires the development of affordable and lightweight intelligent systems capable of assisting users in waste classification while remaining suitable for low-resource environments.

This study therefore proposes the design and implementation of a low-cost object detection system for waste sorting and recycling using lightweight deep learning techniques and publicly available waste datasets. The proposed system is intended to serve as a foundation for future localized adaptations and real world deployment in resource-constrained environments.

1.3 OBJECTIVES OF THE STUDY

1.3.1 GENERAL OBJECTIVE

The main objective of this study is to design and implement a low-cost, human-assisted object detection system for waste sorting and recycling using lightweight deep learning techniques suitable for resource-constrained environments.

1.3.2 SPECIFIC OBJECTIVES

To achieve the general objective, the study will:

- Investigate current waste sorting practices in Cameroon, with particular reference to HYSACAM, Buea, and use the findings to identify and define relevant waste categories for the waste classification system.
- Prepare and preprocess a waste image dataset using publicly available datasets such as Kaggle Datasets, applying domain adaptation techniques such as image augmentation and normalization to improve model robustness, then train and fine-tune a lightweight object detection model (YOLOv8n) using transfer learning, and optimize the trained model through quantization (INT8) for deployment on low-resource devices.
- Design and implement a web-based "WasteDetect" system for real time waste classification using mobile devices, and evaluate the performance of the proposed system using metrics such as mAP@50, precision, recall, inference speed, and model size.

1.4 PROPOSED METHODOLOGY

This study adopts a design and experimental research approach, combining field investigation, machine learning model development, and system implementation.

First, a preliminary field study will be conducted to understand current waste sorting practices at HYSACAM, including waste categories, sorting methods, and operational challenges. The findings from this investigation will help define the waste categories relevant to the proposed system.

Next, a dataset of approximately 10,500 waste images will be prepared using publicly available datasets obtained from Kaggle datasets. The dataset will be divided into training, validation, and testing subsets using appropriate machine learning data splitting techniques.

Data preprocessing will involve domain adaptation techniques, including image augmentation methods such as Gaussian noise, motion blur, brightness variation, and normalization using OpenCV and Albumentations. These preprocessing techniques are intended to improve model robustness under varying environmental conditions.

For model development, a YOLOv8n lightweight object detection architecture will be employed using transfer learning. The model will be trained and fine-tuned on the prepared dataset under controlled experimental conditions. System performance will be evaluated using metrics such as mean Average Precision (mAP@50), precision, recall, inference speed, and model size.

To ensure suitability for low-resource environments, post-training quantization (INT8) will be applied to reduce computational requirements and storage size while maintaining acceptable classification accuracy.

Finally, a web-based application named "WasteDetect" will be developed using Flask. The application will allow users to capture waste images through mobile devices using WebRTC technology and obtain real time waste classification results. An optimized local file system directory structure will be utilized to store raw images assets into categorical folders, while an embedded SQLite database will be used for lightweight offline-first data persistence of text-based audit logs and image file paths. Data visualization tools will support local monitoring and analysis of classified waste categories.

1.5 RESEARCH QUESTIONS

This study seeks to answer the following questions:

- What are the current waste sorting practices in Cameroon, particularly within organizations such as HYSACAM?
- How can object detection models be adapted to improve waste classification under varying environmental conditions?
- To what extent can image augmentation and preprocessing techniques improve the robustness of waste classification models?
- Can a lightweight object detection model achieve acceptable accuracy for waste classification while maintaining low computational cost?
- Can a low-cost, human-assisted waste sorting system provide a practical and usable solution for improving waste sorting efficiency in resource-constrained environments?

1.6 RESEARCH HYPOTHESIS

The study is guided by the following hypotheses:

- A lightweight YOLOv8n object detection model can achieve acceptable waste classification performance while maintaining low computational cost.
- Domain adaptation techniques such as image augmentation and normalization significantly improve the robustness and accuracy of waste classification models.
- Post-training quantization (INT8) can significantly reduce model size and computational requirements without causing major loss in classification accuracy.
- A web-based, human-assisted waste sorting system provides a more practical and cost effective solution for resource-constrained environments compared to fully automated waste sorting systems.

1.7 SIGNIFICANCE OF THE STUDY

This study is significant from technological, environmental, academic, and socio-economic perspectives. The proposed system contributes to the growing field of artificial intelligence and computer vision by demonstrating the application of lightweight object detection models for waste sorting in resource-constrained environments.

From an environmental perspective, the study aims to promote better waste management and recycling practices by assisting users in identifying and sorting waste materials more efficiently. Improved waste sorting can contribute to reducing environmental pollution, minimizing improper waste disposal, and increasing recycling effectiveness.

The study is also significant to waste management stakeholders such as HYSACAM and environmental agencies, as it proposes a low-cost and practical technological solution that can support existing waste management processes without requiring expensive automated infrastructures.

Academically, the research contributes to existing literature on AI-based waste classification systems by exploring the use of lightweight deep learning techniques, transfer learning, and model optimization for low-resource deployment. The study may also serve as a reference for future researchers interested in sustainable technology, computer vision, and smart waste management systems.

Finally, the proposed web-based “WasteDetect” system demonstrates how accessible technologies such as mobile devices and web applications can be integrated with artificial intelligence to provide scalable and user-friendly environmental solutions.

1.8 SCOPE OF THE STUDY

This study focuses on the design and implementation of a low-cost object detection system for waste sorting and recycling using lightweight deep learning techniques.

The study specifically covers:

- Waste classification using image-based object detection techniques.
- Classification of recyclable waste materials into the following categories identified during the field investigation at HYSACAM: Plastic waste, Metal waste, Organic waste, Paper waste, Glass waste.

- The use of publicly available waste datasets such as Kaggle Datasets.
- Data preprocessing and augmentation techniques for improving model robustness.
- Development and optimization of a YOLOv8n-based object detection model.
- Implementation of a web-based waste classification application using Flask and WebRTC.
- Evaluation of the system using performance metrics such as mAP@50, precision, recall, inference speed, and model size.
- Application of lightweight AI techniques suitable for low-resource environments.
- Development of a human-assisted waste sorting prototype intended for resource-constrained settings.

1.9 DELIMITATION OF THE STUDY

This study is limited to the development of a software-based waste classification system and does not include the implementation of physical automated sorting hardware such as robotic arms, conveyor belts, or industrial recycling machines.

The study relies primarily on publicly available datasets due to limitations in time, logistics, and access to large-scale locally collected waste images. Consequently, the system may not fully represent all real world waste conditions specific to Cameroon.

In addition, the study focuses only on image-based waste classification and does not address other aspects of waste management such as waste collection, transportation, disposal logistics, or recycling plant operations.

The proposed system is designed as a prototype intended for academic and experimental purposes and may require further optimization and real world testing before large-scale deployment.

1.10 DEFINITION OF KEYWORDS AND TERMS

- **Artificial Intelligence (AI)**

A branch of computer science that enables machines and computer systems to perform tasks that typically require human intelligence, such as learning, reasoning, and decision-making.

- **Object Detection**

A computer vision technique used to identify and locate objects within an image or video using bounding boxes and class labels.

- **Waste Sorting**

The process of separating waste materials into different categories to facilitate recycling, reuse, or proper disposal.

- **Recycling**

The process of converting waste materials into reusable products to reduce environmental pollution and resource consumption.

- **YOLO (You Only Look Once)**

A real time object detection algorithm that detects and classifies objects in images using a single neural network pass.

- **YOLOv8n**

The lightweight version of the YOLOv8 object detection model optimized for low computational cost and fast inference.

- **Transfer Learning**

A machine learning technique where a pre-trained model is adapted for a new task by fine-tuning it on a different dataset.

- **Domain Adaptation**

Techniques used to improve model performance under varying environmental conditions through preprocessing and augmentation methods.

- **Image Augmentation**

The process of artificially increasing dataset diversity by applying transformations such as rotation, blur, brightness adjustment, and noise.

- **Quantification**

A model optimization technique that reduces computational complexity and model size by converting high-precision numerical values into lower-precision formats such as INT8.

- **mAP (Mean Average Precision)**

A performance metric commonly used in object detection to evaluate how accurately a model detects and classifies objects.

- **WebRTC**

A web technology that enables real time communication and camera access directly through web browsers.

1.11 ORGANIZATION OF THE DISSERTATION

This dissertation is organized into five chapters.

Chapter One: General Introduction

This chapter presents the background and context of the study, problem statement, objectives, proposed methodology, research questions, hypotheses, significance, scope, delimitations, definition of key terms, and organization of the dissertation.

Chapter Two: Literature Review

This chapter reviews existing literature related to waste management systems, artificial intelligence in waste sorting, object detection techniques, deep learning models, datasets, and related works relevant to the study.

Chapter Three: Analysis and Design

This chapter presents the system analysis, requirements specification, dataset preparation, system architecture, model design, and design methodologies used in developing the proposed system.

Chapter Four: Implementation and Results

This chapter discusses the implementation of the proposed system, model training process, evaluation metrics, experimental results, system testing, and discussion of findings.

Chapter Five: Conclusion and Future Work

This chapter summarizes the study, presents conclusions based on the findings, discusses limitations, and provides recommendations and possible directions for future research and improvements.

CHAPTER 2. LITERATURE REVIEW

2.1 INTRODUCTION

Waste management and recycling have become major global concerns due to the rapid increase in municipal solid waste generation and the environmental challenges associated with improper disposal practices. According to the World Bank, global waste generation is expected to increase significantly due to rapid urbanization and population growth [1]. Efficient waste sorting plays a critical role in improving recycling processes, reducing environmental pollution, and promoting sustainable waste management systems [4]. In recent years, artificial intelligence (AI), computer vision, and deep learning techniques have increasingly been applied in waste classification and sorting systems to improve efficiency and automation [2]. However, most intelligent waste sorting systems are designed for developed countries with advanced technological infrastructures, making them difficult to deploy in resource-constrained environments such as Cameroon. Challenges such as poor waste segregation practices, mixed waste collection systems, limited automation, inadequate datasets, and high deployment costs continue to hinder effective recycling processes in many developing regions [4]. This chapter therefore reviews existing literature related to waste management practices, AI-based waste sorting systems, object detection techniques, lightweight deep learning models, image preprocessing and augmentation methods, and related waste classification systems.

2.2 GENERAL CONCEPTS ON INTELLIGENT WASTE SORTING SYSTEMS

2.2.1 WASTE MANAGEMENT AND WASTE SORTING

Waste management refers to the collection, transportation, processing, recycling, and disposal of waste materials in ways that minimize environmental and public health impacts [1]. Waste sorting is one of the most important stages in waste management because it facilitates recycling and reuse by separating waste into categories such as plastic, metal, paper, glass, and organic waste [4]. In many developed countries, waste sorting is performed using automated systems that integrate robotics, conveyor belts, sensors, and AI technologies to improve sorting efficiency and recycling accuracy [8]. These systems often combine computer vision and deep learning models to identify recyclable materials in real time. In contrast, waste management systems in many developing countries remain largely manual and poorly structured [4]. Findings from a field interview conducted at HYSACAM revealed that there is currently no effective institutional waste sorting mechanism implemented by the organization. Instead, some recyclable materials are manually removed by workers for personal resale purposes. Additionally, waste items are often mixed together during collection and later transported to disposal sites located outside urban areas. These findings reinforce observations made by previous studies on waste management challenges in developing countries [4]. The interview further identified plastic, metal, and organic waste as the most relevant waste categories that could potentially be prioritized for sorting operations within the organization.

2.2.2 ARTIFICIAL INTELLIGENCE IN WASTE SORTING

Artificial intelligence has significantly transformed modern waste management systems through the development of automated waste classification and sorting technologies [2]. AI-based waste sorting systems commonly employ machine learning and deep learning algorithms to detect, classify, and separate waste materials automatically. Deep learning models, particularly Convolutional Neural Networks (CNNs), have demonstrated strong performance in image recognition and object detection tasks [9]. According to Zhu et al., [2], computer vision systems can effectively identify and classify waste materials in real time environments using deep learning techniques. Similarly, Yang et al. [8] reported that AI-based waste classification systems can achieve high detection accuracy when trained on sufficiently diverse datasets. Despite these advancements, many AI-based systems rely heavily on expensive computational infrastructures such as GPUs, robotic sorting systems, and industrial automation equipment, making them less practical for deployment in low-resource environments [7].

2.2.3 OBJECT DETECTION MODELS FOR WASTE CLASSIFICATION

Object detection is a computer vision technique used to identify and localize objects within images or videos using bounding boxes and class labels [10]. Modern object detection models are generally categorized into:

- Two-stage detectors (e.g., R-CNN family)
- One-stage detectors (e.g., YOLO and SSD)

Two-stage detectors generally provide high detection accuracy but require higher computational resources and slower inference speed [11]. In contrast, one-stage detectors such as YOLO are optimized for real time detection and computational efficiency [10]. The YOLO (You Only Look Once) family has gained widespread popularity due to its balance between speed and accuracy. According to Cheng [12], YOLOv8 achieves strong object detection performance through deep convolutional architectures while maintaining efficient real time detection capabilities. YOLOv8n, the lightweight variant of YOLOv8, is specifically designed for resource-constrained devices and low-latency applications [12]. However, deeper neural network architectures increase computational complexity and storage requirements, making deployment difficult on mobile and embedded devices. To address these challenges, lightweight optimization techniques such as pruning and quantization are increasingly being applied to reduce model size and computational cost while maintaining acceptable detection accuracy [12].

2.2.4 LIGHTWEIGHT DEEP LEARNING MODELS AND COMPUTATIONAL OPTIMIZATION

The deployment of deep learning models on low-resource devices requires optimization techniques that reduce computational complexity and memory consumption [13]. According to Cheng [12], pruning algorithms remove redundant channels and parameters from convolutional neural networks, thereby improving computational efficiency and reducing model size. Quantization techniques further compress models by converting floating-point operations into lower-precision numerical representations such as INT8, significantly reducing memory usage and inference requirements. Experimental results presented by Cheng [12] showed that combining pruning and INT8 quantization achieved a compression ratio of approximately 64.2% with only minor accuracy loss, demonstrating the suitability of lightweight YOLOv8 models for mobile and embedded deployment environments. Other lightweight neural network architectures such as MobileNet, ShuffleNet, and SqueezeNet have also been developed to improve deployment efficiency on low-resource devices [13]. These lightweight architectures prioritize computational efficiency while maintaining acceptable performance levels.

2.2.5 IMAGE AUGMENTATION AND PREPROCESSING TECHNIQUES

Environmental variability presents a major challenge for object detection systems. Variations in lighting conditions, background clutter, image quality, camera motion, and occlusion can significantly reduce detection accuracy [14]. To improve robustness, researchers commonly apply preprocessing and image augmentation techniques during dataset preparation [15]. Common augmentation techniques include:

- Gaussian noise
- Motion blur

- Brightness variation
- Rotation
- Flipping
- Scaling

These techniques artificially increase dataset diversity and improve model generalization under varying environmental conditions [14]. Normalization techniques are also widely used to standardize image representations before model training [9]. Tools such as OpenCV and Albumentations are frequently employed for preprocessing and augmentation operations. Domain adaptation techniques help object detection models become more resilient to environmental variations and improve real world performance [16].

2.2.6 HUMAN-ASSISTED WASTE SORTING SYSTEMS

Fully automated waste sorting systems often require expensive industrial infrastructures including robotic manipulators, conveyor belts, and sensor arrays [7]. Such systems may be impractical for developing countries due to high acquisition and maintenance costs. Human-assisted waste sorting systems provide a more feasible alternative by integrating AI-based classification tools with human decision-making processes [4]. Instead of fully replacing human workers, these systems support users in identifying and categorizing waste materials more efficiently. Web-based and mobile-assisted waste classification systems are increasingly being explored because they leverage widely accessible technologies such as smartphones and web browsers [2]. Lightweight AI models combined with mobile camera interfaces can therefore provide practical, scalable, and cost effective waste sorting assistance in low-resource environments.

2.3 RELATED WORKS

Several studies have explored the application of deep learning and computer vision techniques in waste classification and recycling systems. Zhu et al., [2] developed a computer vision system capable of detecting and classifying waste materials in urban street environments using deep learning methods. Their system demonstrated the effectiveness of AI-based waste detection under real world conditions but required significant computational resources. Thung and Yang [8] introduced the TrashNet dataset, one of the most widely used benchmark datasets for waste classification research. Although the dataset has contributed significantly to AI-based recycling studies, it contains limited environmental diversity and controlled image conditions, which may affect real world generalization. Cheng [12] proposed a lightweight YOLOv8 optimization approach using pruning and INT8 quantization techniques to reduce computational complexity while maintaining acceptable detection performance. Experimental results showed that the optimized model achieved substantial compression with minimal accuracy degradation, making it suitable for deployment on resource-constrained devices. Ferronato and Torretta [4] discussed waste management challenges in developing countries and emphasized that poor waste segregation, limited technological infrastructure, and inadequate recycling systems continue to hinder sustainable waste management efforts. [17] observed that many intelligent waste sorting systems remain difficult to deploy in developing countries because of high hardware and operational costs associated with automation infrastructures.

Despite the progress achieved by previous studies, several research gaps remain:

- Many systems depend on expensive computational infrastructures
- Existing datasets often lack environmental variability
- Limited focus exists on low-cost deployment in developing countries
- Few systems combine lightweight object detection with human-assisted interaction models

The proposed study seeks to address these gaps by developing a lightweight, low-cost, web-based waste sorting system suitable for resource-constrained environments.

2.4 PARTIAL CONCLUSION

This chapter reviewed existing literature related to waste management systems, AI-based waste classification, object detection models, lightweight deep learning techniques, image preprocessing methods, and human-assisted waste sorting systems. The review revealed that although modern AI-based waste sorting systems achieve high classification performance, many existing solutions remain computationally expensive and unsuitable for deployment in low-resource environments [7]. The literature also highlighted the importance of image augmentation, preprocessing, and lightweight optimization techniques in improving object detection robustness and deployment efficiency [14]. Furthermore, findings from the field investigation conducted at HYSACAM revealed the absence of effective institutional waste sorting mechanisms, reinforcing the need for practical and affordable intelligent waste classification systems. Based on the identified research gaps, the proposed study aims to design and implement a lightweight, human-assisted waste sorting system capable of operating efficiently in resource-constrained environments while improving waste classification and recycling support.

CHAPTER 3. ANALYSIS AND DESIGN

3.1 INTRODUCTION

This chapter presents the analysis of requirements gathered during the brainstorming and field investigation phases, and the design decisions made in response to those requirements. It covers the methodology adopted, the system design supported by requirements specifications and diagrams, the global solution architecture, and the key algorithms chosen; with the reasoning behind each choice.

The central problem is that waste management in Cameroon remains largely manual [17], and the AI-based sorting systems documented in existing literature require expensive infrastructure that is impractical for resource-constrained environments [7]. At the same time, public waste datasets tend to lack the environmental variability needed for a model to perform reliably under the varying lighting and camera conditions typical of mobile capture in Buea[14]. These two gaps; high deployment cost and limited dataset variability, directly shaped every major design decision in this chapter.

3.2 METHODOLOGY

A design and experimental research methodology was adopted. The work was carried out in five sequential phases, described below.

3.2.1 PHASE 1: BRAINSTORMING AND DESIGN CONSIDERATIONS

Before any field work or technical implementation began, a brainstorming phase was carried out to weigh the practical cost and feasibility implications of building an AI-based waste classification system for a resource-constrained context. Five considerations were examined.

- **Implementation cost:** The software resources used to build the model and system are entirely open source (Python, Ultralytics YOLOv8, Flask, React, Albumentations, OpenVINO). The only real monetary cost identified was the cost of hosting the application.
- **Technical cost:** A lightweight model architecture (YOLOv8n) combined with INT8 quantization was chosen specifically to reduce power consumption, memory usage, and

computational complexity, since the target deployment hardware would not include GPUs or industrial computing infrastructure.

- **Deployment cost:** The system needed to run on devices already owned by the target users (ordinary mobile phones) rather than requiring dedicated or specialised hardware. This ruled out any solution that depended on industrial sorting equipment.
- **Maintenance cost:** Because the system is software-only, ongoing maintenance is restricted to the application layer (bug fixes, model updates, server upkeep) rather than the maintenance of physical sorting hardware, which would be considerably more expensive to sustain.
- **Practicality:** The overall goal was to balance real time responsiveness with low operating cost, while keeping the interface usable by citizens with limited technical experience. This consideration directly shaped the choice of a simple mobile-first PWA interface over a more complex native application.

These considerations collectively justified the core architectural direction of the project: a lightweight, quantized object detection model deployed within a low-cost, mobile-accessible web application, rather than a fully automated industrial sorting system. This direction was then validated against real operational needs during the field study described in Phase 2.

3.2.2 PHASE 2: FIELD STUDY AT HYSACAM BUEA

A structured field investigation was conducted at HYSACAM, Buea, to validate the brainstormed design direction against real operational needs and to identify which waste categories the organisation considers most relevant for classification.

24 April 2026: First visit to HYSACAM. Officials could not grant access without a formal authorisation letter from the University.

5 May 2026: Authorisation letter collected from the University of Buea.

8 May 2026: Second visit to HYSACAM. The Regional Secretary accepted the letter but additionally requested a handwritten letter addressed to the organisation and a typed questionnaire. Both were drafted on-site, a process that took three to four hours to ensure the questionnaire was clear and easy for non-technical staff to complete.

13 May 2026: The completed questionnaire was retrieved after two hours of follow-up.

The questionnaire results revealed the following:

- HYSACAM Buea collects waste through three methods: door-to-door collection, container collection, and scheduled kerbside pickup.
- Main operational challenges include frequent vehicle breakdowns and illegal dumping by residents.
- HYSACAM does not currently sort waste. No institutional sorting mechanism exists.
- Officials expressed interest in a system that could assist with sorting into **plastic, metal, and organic** waste (top priorities), with paper and glass as desirable additions. Such a system was seen as beneficial for reducing manual sorting time and lowering landfill volumes.

These findings confirmed the absence of any sorting mechanism at HYSACAM, validated the low-cost design direction set during brainstorming, and provided the operational basis for defining the five target waste categories used throughout the system design.

3.2.3 PHASE 3: DATASET PREPARATION

A publicly available waste image dataset (viswaprakash1990/garbage-detection, Kaggle) was used. The original train/valid/test split was discarded to eliminate pre-existing bias. All 10,464 valid image-annotation pairs were pooled, globally shuffled with a fixed seed (seed = 42) for reproducibility, and re-split into 70% training, 20% validation, and 10% test subsets. The six source classes were remapped to five HYSACAM-aligned target classes. All images were verified for integrity; corrupt files were removed, and images without a label file were assigned empty annotations to register them as background examples in YOLO format.

3.2.4 PHASE 4: MODEL TRAINING AND OPTIMIZATION

A YOLOv8n model was fine-tuned on the prepared dataset using transfer learning from COCO pre-trained weights. Five image augmentation pipelines were applied to the training split only, expanding it from 7,324 to 43,869 images. After training, post-training INT8 quantization was applied via the OpenVINO toolkit. The model was evaluated on both the validation split (for optimisation comparison) and the held-out test split (for final production metrics).

3.2.5 PHASE 5: SYSTEM DESIGN AND IMPLEMENTATION

The optimised INT8 OpenVINO model was extracted and integrated into a Flask backend. The WasteDetect web application was designed and built following the standard Software Development Life Cycle (SDLC), with the INT8 model serving as the AI inference engine.

3.3 SYSTEM DESIGN

3.3.1 SYSTEM REQUIREMENTS

The system requirements were derived from the brainstorming considerations and the HYSACAM field study findings. They are organised into functional requirements (what the system must do) and non-functional requirements (how well the system must perform).

Table 1: Functional Requirements

ID	Requirement	Actor
FR01	Allow users to register with name, email, and password	User
FR02	Allow users to log in and receive a JWT token	User / Admin
FR03	Allow users to capture a waste image in real time via WebRTC	User
FR04	Send the captured image to the Flask backend for classification using YOLOv8n	System
FR05	Return the predicted waste category (Plastic, Metal, Organic, Paper, Glass) with a confidence score	System
FR06	Store the classification result, confidence score, user correction (if any), image URL, and timestamp	System
FR07	Upload the captured image to Cloudinary and store the returned URL in the database	System
FR08	Display to the user how many waste items they have sorted	User

ID	Requirement	Actor
FR09	Display an encouragement message showing environmental impact of the sorting action	User
FR10	Store classification results locally in IndexedDB when offline and sync automatically on reconnection	System
FR11	Allow the admin to view total items sorted for the current day and month	Admin
FR12	Allow the admin to view the percentage of correctly and incorrectly sorted items	Admin
FR13	Auto-generate a brief explanation for each wrong sort and store it in a log	System
FR14	Display admin statistics using bar, pie/doughnut, and line charts	Admin
FR15	Restrict access to the admin dashboard to authorised admin users only	System
FR16	Allow the PWA to be installed on a mobile device like a native app	User

Table 2: Non-Functional Requirements

ID	Requirement	Category
NFR01	Return a classification result within 3 seconds on a standard mobile connection	Performance
NFR02	Function on low-end Android mobile devices with limited RAM	Performance
NFR03	Use the YOLOv8n INT8 quantized model to minimise inference time and memory usage	Performance
NFR04	Use HTTPS for all API communication between React and Flask in production	Security
NFR05	Hash passwords before storage	Security
NFR06	Expire JWT tokens after a defined period and require re-authentication	Security
NFR07	Be usable by a non-technical Cameroonian citizen with minimal training	Usability
NFR08	Provide the UI in English for the MVP phase	Usability
NFR09	Handle poor internet connectivity gracefully using offline sync	Reliability
NFR10	Be deployable on Railway with PostgreSQL without code changes from development	Portability
NFR11	Be deployable on Vercel without environment-specific code changes	Portability

ID	Requirement	Category
NFR12	Log all wrong-sort events with timestamps, user ID, and model prediction	Maintainability

In addition to the requirements above, the system operates under several fixed constraints: the YOLOv8n INT8 model is fixed with no in-app retraining; the system is scoped as a prototype for Buea, Cameroon rather than national deployment; only single-image classification is supported (no video stream processing); admin accounts are created manually in the database with no public registration; image storage relies on Cloudinary's free tier; and SQLite is used only in development, with PostgreSQL used in production.

3.3.2 MUST-HAVE FEATURES (MOSCOW-PRIORITIZATION)

Feature prioritisation followed the Moscow method. The eight Must-Have features below represent the minimum viable scope required to satisfy the functional and non-functional requirements above and were the basis for the system design that follows.

Table 3: Must-Have Features

ID	Feature	Description
M01	User Authentication	Registration, login, JWT management, password hashing, and role assignment.
M02	Waste Classification	Image capture, Cloudinary upload, YOLOv8n inference, confidence routing, and database record saving as one seamless action.
M03	Classification Result Display	The user immediately sees the predicted category and confidence score after classification.
M04	User Stats	The user can view how many waste items they have sorted.
M05	Encouragement and Impact Message	After each classification, the user receives a message on the environmental impact of their action.
M06	Admin Dashboard	The admin views total items sorted daily and monthly, with a high/low confidence breakdown.
M07	PWA Installability	The user can install WasteDetect on a mobile device like a native app, directly from the browser.
M08	Offline Sorting with Auto Sync	Captured images are queued locally when offline and auto-submitted once connectivity returns.

3.3.3 TECHNOLOGY STACK

WasteDetect is a client-server Progressive Web Application (PWA). Every technology choice was guided by the constraint of low cost and suitability for resource-constrained deployment, consistent with the brainstormed cost considerations in Section 3.2.1. Table 4 presents the full stack.

Table 4: WasteDetect Technology Stack

Category	Technology	Purpose
Frontend	React.js	Component-based UI for the user interface and admin dashboard
Camera Access	WebRTC	Real time camera access on mobile devices for image capture
Data Charts	Chart.js	Bar, pie/doughnut, and line charts for dashboards
Offline Sync	IndexedDB + Service Worker	Queues images locally when offline; auto-uploads on reconnection
Backend	Flask (Python)	Lightweight web framework; integrates natively with YOLOv8n
Authentication	Flask-JWT-Extended + JWT	Secure login and token-based session management
Access Control	RBAC (role field in user table)	Restricts admin-only routes from standard users
Wrong Sort Log	Confidence threshold + log	Flags low-confidence predictions; logs corrections
Dev Database	SQLite	Lightweight database for local development only
Prod Database	PostgreSQL (Railway)	Persistent relational database for production
Image Storage	Cloudinary	Cloud storage for waste images; returns permanent URLs
Frontend Hosting	Vercel (free)	Hosts the React PWA with automatic GitHub deployments
Backend Hosting	Railway (~$5/month)	Hosts Flask backend and PostgreSQL

3.3.4 SYSTEM ARCHITECTURE DIAGRAM

The system is organised into three layers. The Frontend (hosted on Vercel) contains the React UI with the WebRTC camera interface and Chart.js dashboards, a Service Worker for offline support, and IndexedDB for local image queuing. The Backend (hosted on Railway) contains the Flask API, JWT Auth Service, YOLOv8n INT8 inference engine, and Mismatch Logger. The Storage layer contains PostgreSQL for structured records and Cloudinary for waste image file storage.

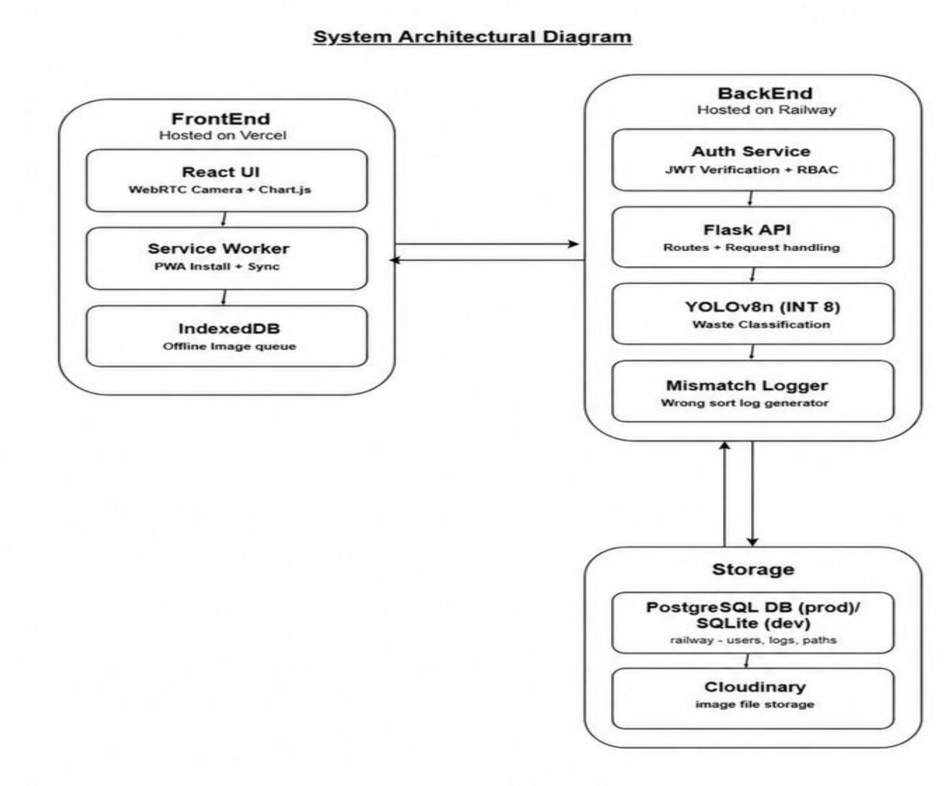

Figure 1: WasteDetect System Architecture Diagram

3.3.5 CONTEXT DIAGRAM

The Context Diagram shows two external entities (user and admin) and two other services (Cloudinary and PostgreSQL/SQLite) that interact with WasteDetect. The Citizen (User) submits waste images and receives a classification result, bin placement instruction, and personal impact message. The admin (HYSACAM staff) accesses classification statistics and audit logs. Cloudinary stores waste images and returns permanent URLs. PostgreSQL/SQLite persists all classification records.

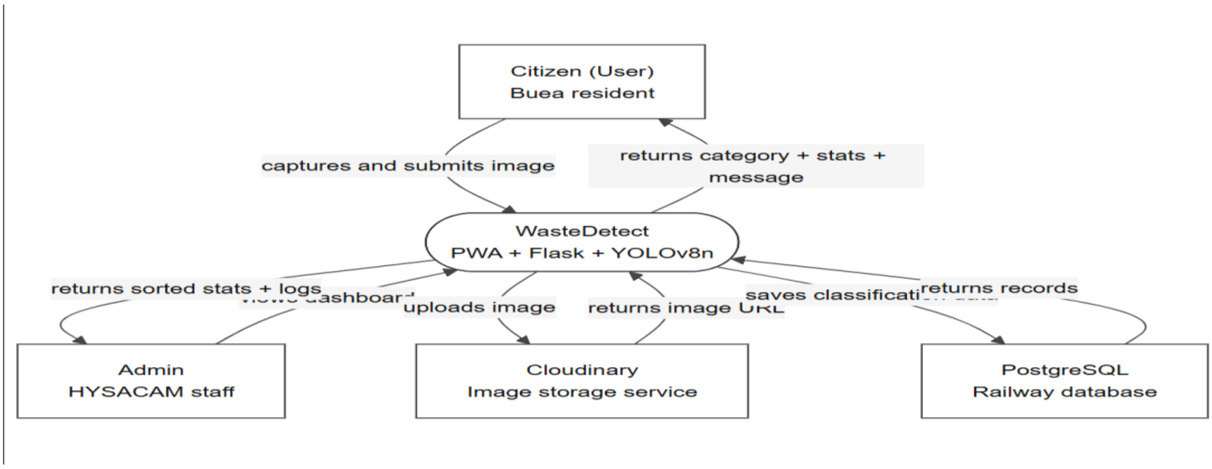

Figure 2: WasteDetect Context Diagram

3.3.6 USE CASE DIAGRAM

Two primary actors interact with the system: the user (citizen) and the admin (HYSACAM official).

User use cases: Register/Log in; Capture waste image; Receive classification result (system-triggered); View personal stats and impact message; Offline image queue and background sync.

Admin-only use cases: View admin dashboard; View sort charts; View mismatch and confidence logs; Manage system data.

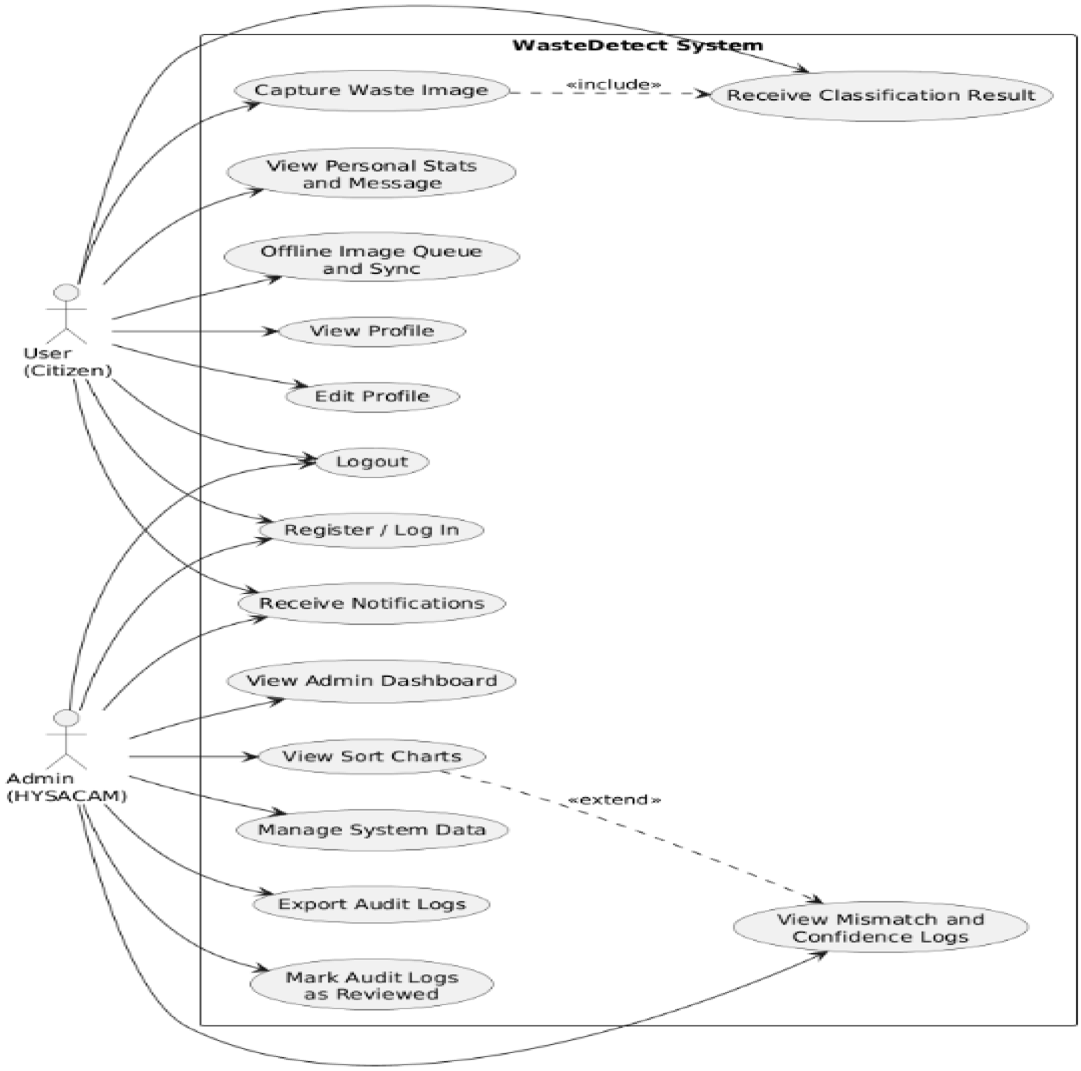

Figure 3: WasteDetect Use Case Diagram

3.3.7 ACTIVITY DIAGRAM

The Activity Diagram describes the flow of a single waste classification session. Two paths exist. In the online path: the user logs in, captures a waste image, and submits it to the Flask API; the Auth Service verifies the JWT; the image is uploaded to Cloudinary and passed to the YOLOv8n INT8 engine; the result is evaluated against a confidence threshold (50%); a classification record is saved to PostgreSQL; and the category, confidence score, and bin placement instruction are returned to the user. In the offline path: the image is stored in IndexedDB; the Service Worker monitors connectivity and automatically submits the queued image once the connection is restored.

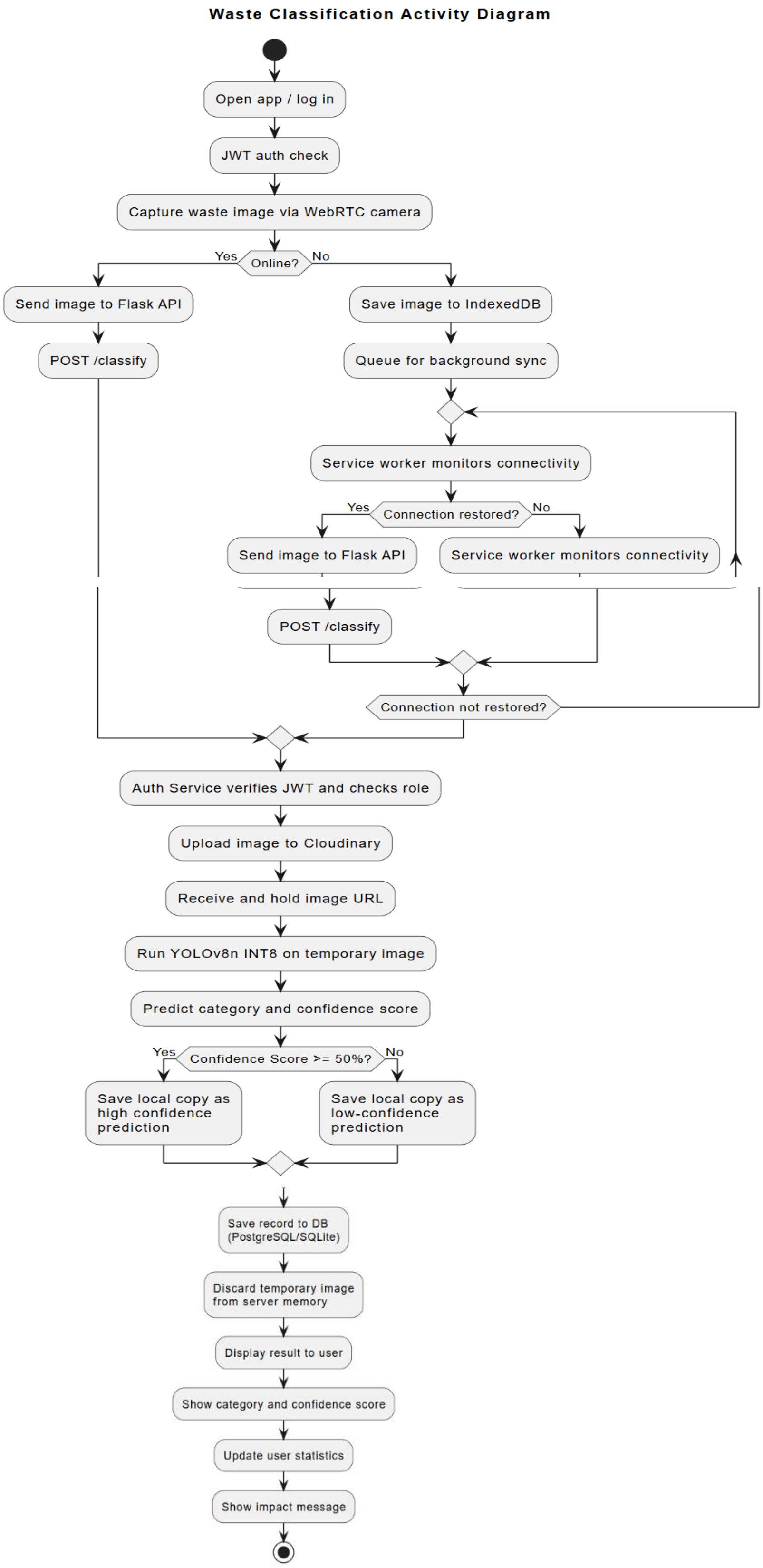

Figure 4: WasteDetect Activity Diagram

3.3.8 ENTITY RELATIONSHIP DIAGRAM (ERD)

The database comprises four tables. USERS stores account details and role assignments. CLASSIFICATIONS records every classification event linked to a user. LOWCONFIDENCEAUDITLOGS stores the auto-generated explanation and admin review status for any classification flagged as low-confidence. MONTHLYSTATS tracks each user's total sorted items per calendar month. Table 5 summarises the key fields, and Figure 5 shows the full ERD.

Table 5: Database Tables and Key Fields

Table	Key Fields
USERS	id (PK), FullName, Email, PasswordHash, Role, CreatedAt, UpdatedAt
CLASSIFICATIONS	id (PK), UserId (FK), ImageUrl, PredictedCategory, ConfidenceScore, IsLowConfidence, CapturedAt
LOWCONFIDENCEAUDITLOGS	id (PK), Classification_id (FK), AutoGeneratedExplanation, ReviewedByAdmin
MONTHLYSTATS	id (PK), UserId (FK), Year, Month, TotalSorted, UpdatedAt

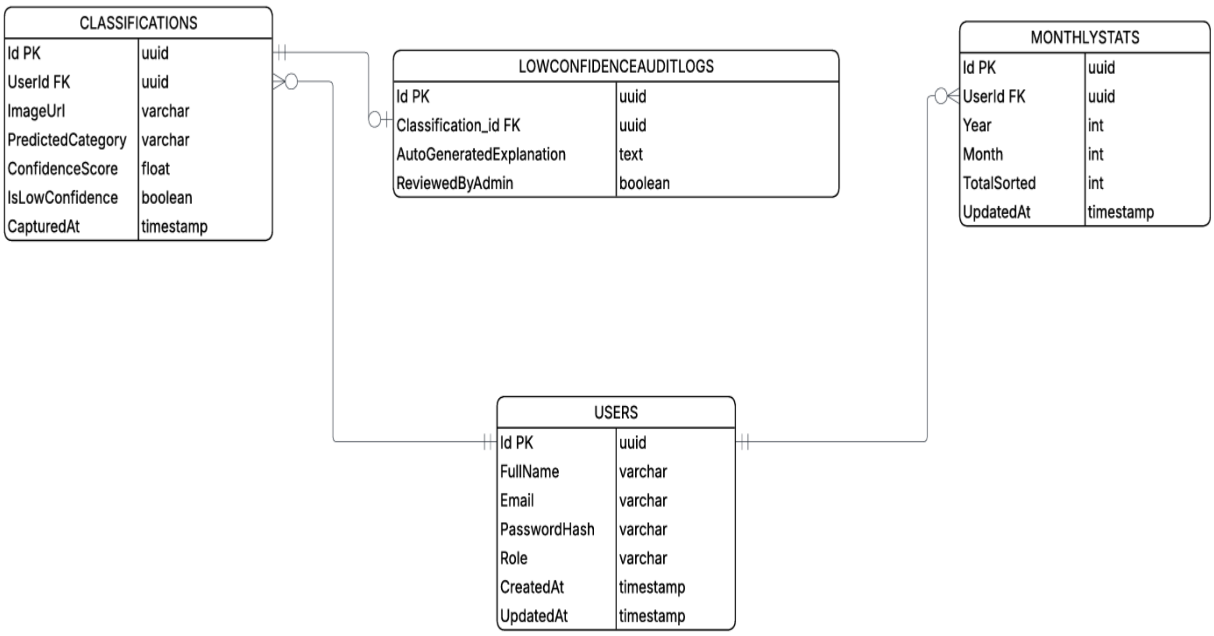

Figure 5: WasteDetect Entity Relationship Diagram

3.4 GLOBAL ARCHITECTURE OF THE SOLUTION

The solution is built around two interconnected pipelines that converge at deployment.

Machine Learning Pipeline: Raw Kaggle images are pooled, shuffled, and re-split (70/20/10). Classes are remapped from six to five. The training split is augmented fivefold (7,324 → 43,869 images). YOLOv8n is fine-tuned via transfer learning, evaluated in FP32, then quantized to INT8 via OpenVINO and evaluated on the held-out test set. The final model is extracted as an OpenVINO IR file (.xml + .bin).

Web Application Pipeline: The React PWA (Vercel) communicates with the Flask REST API (Railway) over HTTPS. A classification request is authenticated via JWT, the image is passed to the YOLOv8n INT8 engine, the result is saved to PostgreSQL and the image to Cloudinary, and the category, confidence score, and bin placement instruction are returned to the user's device.

Note: When the device has no connection, the Service Worker stores the image in IndexedDB and registers a Background Sync task. The image is submitted automatically once connectivity is restored. A visible status banner informs the user at all times.

3.5 KEY ALGORITHMS AND DESIGN CHOICES

3.5.1 YOLOv8n – OBJECT DETECTION MODEL

YOLOv8n (You Only Look Once, nano variant) detects and classifies objects in a single forward pass through a convolutional neural network, without a separate region proposal step[10]. Its anchor-free detection head and reduced parameter count make it significantly faster and more memory-efficient than larger architectures, making it suitable for deployment on mobile devices and low-cost hardware[12]. This directly addresses the technical cost consideration raised during brainstorming.

3.5.2 TRANSFER LEARNING

Rather than training from random initialisation, YOLOv8n was initialised with weights pre-trained on the COCO dataset. This gives the model a strong visual foundation from the start.

Fine-tuning on the 43,869-image waste dataset then specialises the model for the five target categories at a fraction of the data and compute cost that full training would require[9], [12].

3.5.3 CLASS REMAPPING

The source dataset's six classes were consolidated into five target classes aligned with HYSACAM's operational categories. CARDBOARD and PAPER were merged into a single paper class, since both are cellulose-based recyclables sorted identically in practice. BIODEGRADABLE was renamed to organic to match field study terminology. Table 6 shows the full remapping.

Table 6: Class Remapping - Source Dataset to Target Classes

Original Class	Original ID	Target Class	Target ID
BIODEGRADABLE	0	organic	4
CARDBOARD	1	paper	1
GLASS	2	glass	2
METAL	3	metal	3
PAPER	4	paper	1
PLASTIC	5	plastic	0

3.5.4 IMAGE AUGMENTATION

Five Albumentations pipelines were applied to the training split only to simulate the environmental variability that a mobile camera is likely to encounter in Buea. The validation and test splits were kept unmodified to preserve the integrity of evaluation metrics[15]. This directly compensates for the dataset variability gap identified in the literature. Table 7 describes the five pipelines.

Table 7: Image Augmentation Pipelines

Pipeline	Transformations Applied	Condition Simulated
Structural	Horizontal/vertical flip, scale (±20%), rotation (±30°)	Camera angle and distance variations
Sensor noise	Motion blur (kernel 3–7 px), Gaussian noise	Camera shake and noise on low-end mobile devices
Morning light	Colour jitter (brightness ±15%, contrast ±10%) + mild RGB shift	Early-morning warm outdoor lighting

Pipeline	Transformations Applied	Condition Simulated
Afternoon light	Colour jitter (brightness ±30%, contrast ±20%)	Bright mid-day light with high contrast
Evening / dark	Brightness reduction (50–70%), reduced saturation, RGB shift	Dim indoor or evening environments

3.5.5 POST-TRAINING INT8 QUANTIZATION

After training, the FP32 model was exported to OpenVINO INT8 format. Quantization converts 32-bit floating-point weights to 8-bit integers, reducing model size and speeding up inference without re-training. The validation split served as calibration data. [12] demonstrated that this approach achieves approximately 64% compression with only minor accuracy loss, directly supporting the technical and deployment cost considerations identified during brainstorming.

3.5.6 JWT AUTHENTICATION AND ROLE-BASED ACCESS CONTROL

On login, the server issues a signed JWT containing the user's identity and role. Every API request carries this token, which the Auth Service verifies before granting access. Role-Based Access Control (RBAC) is enforced by checking the role field in the user record, ensuring standard users cannot reach administrative endpoints.

3.6 PARTIAL CONCLUSION

This chapter presented the analysis and design of the WasteDetect system. The brainstorming phase identified five cost and practicality considerations that set the overall design direction, and the subsequent field investigation at HYSACAM validated that direction while establishing the five target waste categories. The functional and non-functional requirements, together with the prioritised Must-Have features, defined the minimum scope for the system, and each design decision; YOLOv8n for lightweight detection, transfer learning to reduce training data requirements, five pipeline augmentation to address dataset variability, INT8 quantization to reduce deployment cost, and a PWA with offline support for connectivity-limited environments directly responds to an identified gap, requirement, or operational constraint.

Five system diagrams were presented: the System Architectural Diagram, Context Diagram, Use Case Diagram, Activity Diagram, and Entity Relationship Diagram. Together with the requirements and feature specifications, they define what WasteDetect does, how its components interact, and how its data is structured. Chapter 4 will present the implementation details, model training results, system screenshots, and evaluation of the solution against the study objectives.

CHAPTER 4. IMPLEMENTATION (or REALIZATION) AND RESULTS

4.1 INTRODUCTION

Chapter 3 presented the analysis and design of the WasteDetect system, including the system requirements, architecture, and the methodology chosen to meet the study objectives. This chapter presents the practical realisation of that design. It describes the tools and materials used, the implementation process followed for both the machine learning model and the web application, the results obtained from model training and evaluation, and a final assessment of how well the completed system satisfies the objectives stated in Chapter 1.

User Interface screens and result outputs are used throughout this chapter to support the explanations, since a working system is best understood by seeing it in action.

4.2 TOOLS AND MATERIALS USED

Table 8 summarises the tools, frameworks, and materials used across the machine learning and web application components of the project.

Table 8: Tools and Materials Used

Category	Tool / Material	Purpose
ML Development	Kaggle Notebooks (GPU runtime, Tesla T4 x2)	Cloud environment used for dataset preparation, model training, and evaluation
ML Framework	Ultralytics YOLOv8 (v8.4.58), PyTorch	Model definition, training loop, and validation
Optimization	OpenVINO Toolkit	Post-training INT8 quantization and export of the deployment model
Data Augmentation	Albumentations	Applying the 5 augmentation pipelines to the training split
Dataset Source	Kaggle (viswaprakash1990/garbage-detection)	Source of the 10,464 waste images used for training and evaluation

Category	Tool / Material	Purpose
Backend	Flask (Python), Python 3.12	REST API server and AI inference host
Frontend	React.js, Vite	Progressive Web Application user interface
Authentication	Flask-JWT-Extended	Token-based user session management
Database	SQLite (dev), PostgreSQL (prod, Railway)	Persistent storage of user, classification, and stats records
Image Storage	Cloudinary	Cloud storage and retrieval of captured waste images
Version Control	Git / GitHub	Source code management and Vercel/Railway deployment integration
Hardware (test)	Mid-range Android smartphone, laptop browser	Devices used to test the WebRTC camera capture and PWA installation

4.3 DESCRIPTION OF THE IMPLEMENTATION PROCESS

4.3.1 MACHINE LEARNING MODEL IMPLEMENTATION

The model was implemented and trained entirely within a Kaggle Notebook environment, using a GPU runtime with two NVIDIA Tesla T4 GPUs. The implementation followed the dataset preparation steps described in Chapter 3 (pooling, re-splitting, class remapping, and sanitation), followed by training-split augmentation, model fine-tuning, and post-training quantization.

The YOLOv8n model was fine-tuned using the Ultralytics training API, starting from COCO pre-trained weights. Table 9 lists the exact hyperparameters passed during training.

Table 9: YOLOv8n Training Hyperparameters

Hyperparameter	Value
Pre-trained weights	yolov8n.pt (COCO dataset)
Epochs	30

Hyperparameter	Value
Input image size	640 × 640 pixels
Batch size	32
Data loading workers	4
Random seed	42
Training hardware	2 × NVIDIA Tesla T4 GPU (Kaggle Notebook environment)
Model checkpoint saved	best.pt (highest validation mAP@50 across all epochs)

After the 30-epoch training run completed, the checkpoint with the best validation accuracy (mAP@50) was automatically saved as best.pt. This file was then reloaded and evaluated on the validation split to obtain the FP32 baseline metrics, after which it was exported to OpenVINO INT8 format using the validation split as the calibration dataset. The full procedure, from raw dataset to final quantized model was executed inside a single, reproducible Kaggle Notebook.

4.3.2 BACKEND IMPLEMENTATION

The Flask backend exposes a REST API that wraps the trained INT8 model as an inference service. On startup, the backend loads the OpenVINO INT8 model into memory once, so that subsequent classification requests do not incur model-loading delay. Incoming requests to the classification endpoint are authenticated via JWT, the submitted image is run through the model, and the predicted category, confidence score, and bin placement instruction are returned as a JSON response. Classification records are written to the database, and the original image is forwarded to Cloudinary for permanent storage.

4.3.3 FRONTEND IMPLEMENTATION

The frontend is a React.js Progressive Web Application built with Vite. It implements the WebRTC camera capture interface, the post-classification result screens, the user stats and impact pages, and user profile page. The Service Worker and IndexedDB integration provide offline image queuing, automatically submitting any queued classification once the device reconnects to the internet.

4.3.4 RUNNING THE SYSTEM LOCALLY

The WasteDetect system consists of two independently run servers; a backend API and a frontend client which must both be running for the full system to function during local development or testing.

Backend: After activating the Python virtual environment (venv) at the backend directory level, the server is started with the command "python run.py". The backend then runs at http://localhost:5000.

Frontend: At the frontend directory level, the development server is started with the command "npm run dev". The frontend then runs at http://localhost:5173 and communicates with the backend over the REST API.

Once both servers are running, the application is accessed through a web browser at the frontend address. On a mobile device, the PWA can additionally be installed directly from the browser for a native-app-like experience.

4.4 PRESENTATION AND INTERPRETATION OF RESULTS

4.4.1 MODEL EVALUATION RESULTS

This section presents the model performance results obtained at each stage of evaluation: the FP32 baseline, the INT8 optimised model on the validation split, and the final INT8 model on the unseen test split.

FP32 Baseline (Validation Split): After training, the FP32 PyTorch model was evaluated on the validation split. Figure 6 shows the recorded baseline metrics.

```
Results saved to /kaggle/working/runs/detect/val

=== BASELINE METRICS (FP32 PyTorch) ===
 -> mAP@50   : 0.6401
 -> Precision: 0.7102
 -> Recall   : 0.5670
 -> Latency  : 3.76 ms / image
 -> File Size: 17.52 MB
```

Figure 6: FP32 Baseline Metrics on the Validation Split

The FP32 model achieved an accuracy (mAP@50) of 0.6401, a precision of 0.7102, and a recall of 0.5670, with an average inference latency of 3.76 ms per image and a file size of 17.52 MB. Figure 7 breaks this result down per waste category.

```
[RECOVERY 1/3] Re-calculating Baseline FP32 Performance Metrics...
Ultralytics 8.4.58 🚀 Python-3.12.13 torch-2.10.0+cu128 CUDA:0 (Tesla T4, 14913MiB)
                                                        CUDA:1 (Tesla T4, 14913MiB)
Model summary (fused): 73 layers, 3,006,623 parameters, 0 gradients
val: Fast image access ✅ (ping: 0.0±0.0 ms, read: 438.0±129.5 MB/s, size: 19.3 KB)
val: Scanning /content/mixed_dataset/labels/val.cache... 2092 images, 0 backgrounds, 0 corrup
t: 100% ━━━━━━━━━━━━ 2092/2092 675.0Mit/s 0.0s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 131/131 6.5it/s 20.2s0.1s
                   all       2092      14089       0.71      0.567       0.64      0.434
               plastic        302       1221      0.691      0.534      0.625      0.405
                 paper        651       1870      0.707      0.511      0.606      0.439
                 glass        554       1594      0.757      0.629      0.688      0.527
                 metal        377       1112      0.671       0.66      0.684      0.469
               organic        426       8292      0.725      0.502      0.597       0.33
Speed: 0.9ms preprocess, 3.9ms inference, 0.0ms loss, 1.6ms postprocess per image
Results saved to /kaggle/working/runs/detect/val-3
```

Figure 7: FP32 Per-Class Validation Results

The per-class breakdown shows that glass (mAP@50 = 0.688) and metal (mAP@50 = 0.684) were the best-performing categories, while paper (mAP@50 = 0.606) and organic (mAP@50 = 0.597) were comparatively harder for the model to classify, likely due to the higher visual variability within these categories in the source dataset.

INT8 Optimised Model (Validation Split): After post-training INT8 quantization, the model was re-evaluated on the same validation split using the OpenVINO runtime. Figure 8 shows the result.

```
Results saved to /kaggle/working/runs/detect/val-2

=== OPTIMIZED METRICS (INT8 OpenVINO) ===
 -> mAP@50    : 0.6309
 -> Precision: 0.7028
 -> Recall    : 0.5594
 -> Latency   : 41.83 ms / image
 -> File Size: 3.00 MB
```

Figure 8: INT8 Optimised Metrics on the Validation Split

Figure 9 shows the full per-class breakdown for the INT8 model on the same validation split.

```
[RECOVERY 2/3] Re-calculating Optimized INT8 OpenVINO Performance Metrics...
WARNING ⚠️ Unable to automatically guess model task, assuming 'task=detect'. Explicitly define
task for your model, i.e. 'task=detect', 'segment', 'classify','pose' or 'obb'.
Ultralytics 8.4.58 🚀 Python-3.12.13 torch-2.10.0+cu128 CUDA:0 (Tesla T4, 14913MiB)
                                                       CUDA:1 (Tesla T4, 14913MiB)
Loading /kaggle/working/weights/best_int8_openvino_model/ for OpenVINO inference...
Using OpenVINO LATENCY mode for batch=1 inference on CPU...
Setting batch=1 input of shape (1, 3, 640, 640)
val: Fast image access ✅ (ping: 0.0±0.0 ms, read: 710.3±320.8 MB/s, size: 19.8 KB)
val: Scanning /content/mixed_dataset/labels/val.cache... 2092 images, 0 backgrounds, 0 corrup
t: 100% ━━━━━━━━━━━━ 2092/2092 731.2Mit/s 0.0s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 2092/2092 18.4it/s 1:54<0.1ss
                   all       2092      14089      0.703      0.559      0.631      0.428
               plastic        302       1221      0.711      0.515      0.614      0.392
                 paper        651       1870      0.696      0.511      0.597      0.433
                 glass        554       1594      0.739      0.635      0.685      0.524
                 metal        377       1112      0.638      0.648      0.675      0.467
               organic        426       8292      0.731      0.488      0.585      0.323
Speed: 0.9ms preprocess, 41.4ms inference, 0.0ms loss, 6.4ms postprocess per image
Results saved to /kaggle/working/runs/detect/val-4
```

Figure 9: INT8 Per-Class Validation Results

Comparative Summary (FP32 vs INT8): Figure 10 presents a direct, side-by-side comparison of the FP32 baseline and the INT8 optimised model across all five-evaluation metrics.

```
================= OPTIMIZATION METRIC SUMMARY REPORT =================
 Metric Category       | Baseline (FP32 PyTorch)  | Optimized (INT8 OpenVINO)
 --------------------------------------------------------------------
 Model Storage Footprint| 17.52 MB                 | 3.43 MB
 Execution Speed (CPU)  | 3.91 ms/img             | 41.37 ms/img
 Accuracy (mAP @ 0.5)   | 0.6401                  | 0.6309
 Model Detection Precision| 0.7102                 | 0.7028
 Model Detection Recall | 0.5670                  | 0.5594
======================================================================
```

Figure 10: FP32 vs INT8 Optimisation Summary

Table 10 reproduces this per-class comparison in tabular form for easier reference, and Table 11 summarises the overall tradeoff between the two model formats.

Table 10: FP32 vs INT8   Per-Class Comparison (Validation Split)

Class	Number of Images	Precision (FP32)	Recall (FP32)	mAP50 (FP32)	Precision (INT8)	Recall (INT8)	mAP50 (INT8)
All (avg.)	2,092	0.710	0.567	0.640	0.703	0.559	0.631
Plastic	302	0.691	0.534	0.625	0.711	0.515	0.614
Paper	651	0.707	0.511	0.606	0.696	0.511	0.597
Glass	554	0.757	0.629	0.688	0.739	0.635	0.685
Metal	377	0.671	0.660	0.684	0.638	0.648	0.675
Organic	426	0.725	0.502	0.597	0.731	0.488	0.585

Table 11: FP32 vs INT8   Overall Optimisation Summary

Metric	FP32 (PyTorch)	INT8 (OpenVINO)	Change
Model size	17.52 MB	3.00 MB	82.9% smaller
Inference speed	3.76 ms/img	41.83 ms/img	slower on CPU (GPU vs CPU tradeoff, see note below)
mAP@50	0.6401	0.6309	1.4% (negligible)
Precision	0.7102	0.7028	1.0% (negligible)
Recall	0.5670	0.5594	1.3% (negligible)

Quantization reduced the model's file size by 82.9% (from 17.52 MB to 3.00 MB) while reducing accuracy (mAP@50) by only 1.4 percentage; points a favourable tradeoff for a system intended to run on low-cost, storage-constrained devices.
Note: The increase in measured inference latency (3.76 ms to 41.83 ms) is not a weakness of quantization itself; rather, it reflects a hardware difference between the two measurements. The FP32 baseline was benchmarked on a GPU during the Kaggle training session, whereas the

OpenVINO INT8 runtime used in this measurement executed on CPU. In real world deployment, where the target devices (ordinary mobile phones and browsers) do not have a dedicated GPU, the INT8 model remains the more practical and deployable option specifically because it was designed and optimised for CPU inference.

Final Production Evaluation (Held-Out Test Split): The INT8 model was finally evaluated on the test split (1,048 images); that were never used during training or validation to obtain an unbiased estimate of real world performance. This evaluation was explicitly run on a CPU (Intel Xeon @ 2.00GHz) to reflect realistic deployment conditions. Figure 11 shows the result.

```
[STEP 7] Final Model Deployment Exam (Evaluating INT8 Model on Unseen TEST Split)...
Ultralytics 8.4.58 🚀 Python-3.12.13 torch-2.10.0+cu128 CPU (Intel Xeon CPU @ 2.00GHz)
Loading /kaggle/working/weights/best_int8_openvino_model/ for OpenVINO inference...
Using OpenVINO LATENCY mode for batch=1 inference on CPU...
Setting batch=1 input of shape (1, 3, 640, 640)
val: Fast image access ✅ (ping: 0.0±0.0 ms, read: 427.7±44.7 MB/s, size: 13.6 KB)
val: Scanning /content/mixed_dataset/labels/test... 1048 images, 0 backgrounds, 0 corrupt: 100%
━━━━━━━━━━━━ 1048/1048 1.3Kit/s 0.8s0.0s
val: New cache created: /content/mixed_dataset/labels/test.cache
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 1048/1048 18.1it/s 57.9s<0.0s
                   all       1048       6909      0.683      0.532      0.598      0.404
               plastic        149        695      0.614      0.376      0.472      0.281
                 paper        322        672      0.713      0.594      0.673      0.534
                 glass        266        848      0.727      0.546      0.597      0.446
                 metal        186        564      0.608      0.678      0.671      0.464
               organic        223       4130      0.756      0.466      0.578      0.293
Speed: 0.8ms preprocess, 40.8ms inference, 0.0ms loss, 7.9ms postprocess per image
Results saved to /kaggle/working/runs/detect/val-5

=== FINAL PRODUCTION DEPLOYMENT METRICS (Unseen Test Set) ===
 -> Final Test mAP@50   : 0.5982
 -> Final Test Precision: 0.6834
 -> Final Test Recall   : 0.5321
```

Figure 11: Final INT8 Model Evaluation on the Held-Out-Test Split

Table 12 summarises the final production metrics.

Table 12: Final INT8 Model - Test Split Results

Metric	**Value (Held-out Test Split, INT8 Model)**
Test set size	1,048 images (10% of total dataset, never seen during training or validation)
Final mAP@50	0.5982

Metric	Value (Held-out Test Split, INT8 Model)
Final Precision	0.6834
Final Recall	0.5321
Inference speed	40.8 ms/image (CPU, Intel Xeon @ 2.00GHz)

The final test accuracy (mAP@50) of 0.5982 is close to the validation accuracy (mAP@50) of 0.6309, indicating that the model generalises reasonably well to unseen data rather than overfitting to the validation split. The small drop in performance between validation and test is expected and typical, since the test split includes some category combinations (such as the more visually diverse plastic class) that are harder to classify consistently.

4.4.2 BENCHMARK COMPARISON

To contextualise these results, Table 13 compares the final accuracy (mAP@50) achieved by WasteDetect's INT8 model against accuracy (mAP@50) values reported by other YOLOv8-based waste detection studies in the literature.

Table 13: Benchmark Comparison with Related YOLOv8-Based Waste Detection Studies

Study / System	Model	mAP@50	Notes
WasteDetect (this study)	YOLOv8n + INT8	0.598	Held-out test split; 5 classes; optimised for low-cost CPU deployment
Web-Based YOLOv8 Waste Detection [18]	YOLOv8s	0.621	Hold-out test set; comparable single-stage detector, larger variant
A comparison of YOLO models [19]	YOLOv8	0.547	WaRP dataset; standard (non-quantized) YOLOv8
MRS-YOLO [20]	Modified YOLOv8	≈ 0.63–0.65	10-category dataset; uses architectural enhancements beyond base YOLOv8n

WasteDetect's final test accuracy (mAP@50) of 0.598 compares favourably against several published YOLOv8-based waste detection systems, including a study reporting 0.547 accuracy (mAP@50) on the WaRP. It is marginally below the 0.621 accuracy (mAP@50) reported by a YOLOv8s-based system; an expected outcome, since YOLOv8s is a larger, heavier model than the YOLOv8n variant used in this study, and WasteDetect deliberately traded a small amount of accuracy for a much smaller, faster, and cheaper to deploy model. This confirms that the lightweight, quantized approach adopted in this study achieves accuracy broadly comparable to existing YOLOv8-based waste detection research, while remaining substantially cheaper to deploy.

4.4.3 SYSTEM INTERFACE SCREENS

The following screens illustrate the completed WasteDetect Progressive Web Application as experienced by a user, from first opening the app through to viewing their personal sorting impact.

Figure 12: WasteDetect Welcome Page

The Welcome page is the first screen a new user sees. It introduces the purpose of the application and provides clear entry points to sign up (Get started) or Sign in.

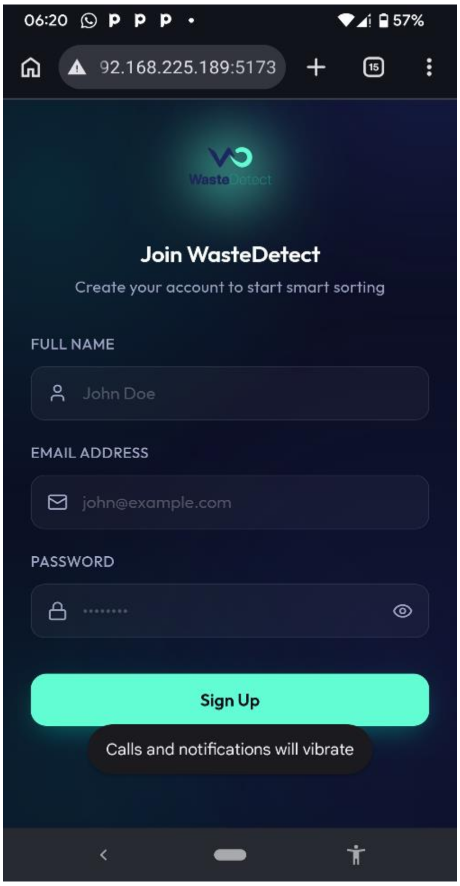

Figure 13: WasteDetect Sign-Up Page

The Sign-Up page collects the user's full name, email, and password to create a new account. When ever all these fields are provided, a user account is created, but if any field is missing, an appropriate error is returned to the user.

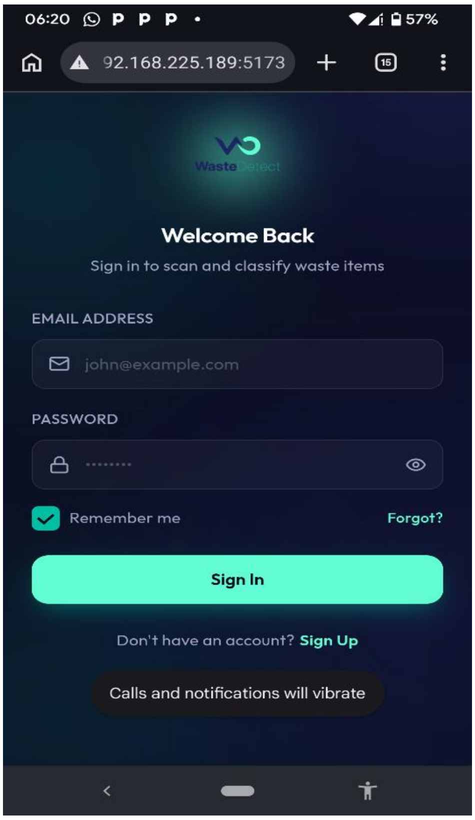

Figure 14: WasteDetect Login Page

The Login page authenticates returning users by collecting their email and password. For successful authentication, the user email address and password must match the one used during registration, otherwise, the authentication will be unsuccessful returning an error to the user.

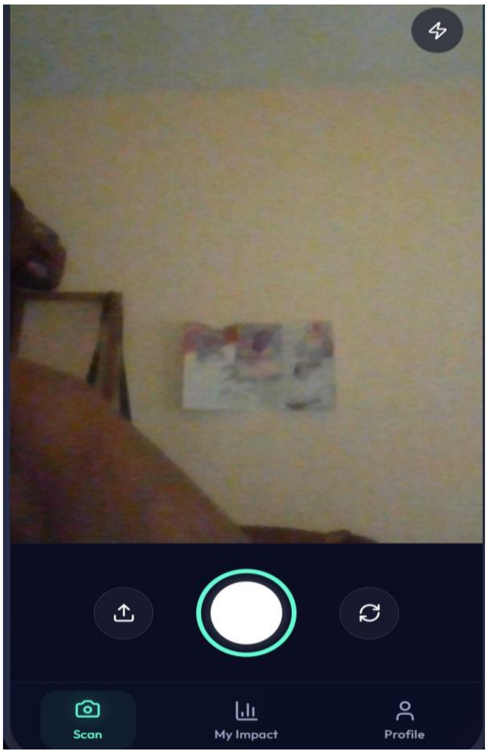

Figure 15: WasteDetect Scan Page

The Scan page provides the WebRTC powered live camera view used to capture a waste image for classification. After capture, the classification result, bin placement instruction, and confidence score are displayed to the user. This page also permits the user to upload waste images directly from his device’s memory for classification.

Figure 16: WasteDetect My Impact Page

The My Impact page displays the user's personal sorting statistics, including total items sorted and daily streak. Users can filter recent sorting activity by waste category, with each entry showing the item, confidence score and timestamp.

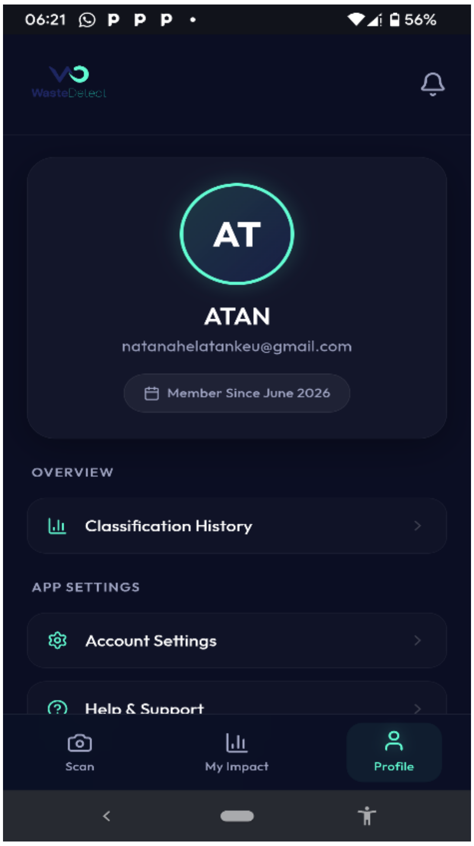

Figure 17: WasteDetect Profile Page

The Profile page displays the logged-in user's account details; name, email, and membership date. Provides navigation to Classification History, Account Settings, and Help & Support for managing personal data and app preferences.

4.5 EVALUATION OF THE SOLUTION

4.5.1 EVALUATION AGAINST STUDY OBJECTIVES

Chapter 1 set out a general objective of designing and implementing a low-cost, human-assisted object detection system for waste sorting suitable for resource-constrained environments, supported by eight specific objectives. The results presented in this chapter demonstrate that each of these objectives has been met: a lightweight YOLOv8n model was successfully trained and fine-tuned via transfer learning; image augmentation was applied to improve robustness; the model was optimised through INT8 quantization with only a small accuracy tradeoff; and the resulting model was integrated into a working, deployable web-based system that classifies waste images in real time and tells the user where to place each item.

4.5.2 LOW-COST COMPARISON WITH EXISTING SOLUTIONS

A central claim of this study is that WasteDetect achieves meaningful waste classification accuracy at a fraction of the cost of conventional AI-based sorting systems. Table 14 compares WasteDetect against typical industrial AI sorting systems across the five cost dimensions identified during the brainstorming phase in Chapter 3.

Table 14: Low-Cost Comparison   WasteDetect vs. Typical Industrial AI Sorting Systems

Cost Dimension	Typical Industrial AI Sorting System	WasteDetect (Proposed System)
Implementation cost	Proprietary computer vision software licences; custom robotics integration	Entirely open source software stack (Python, YOLOv8, Flask, React, OpenVINO); only cost is ~$5/month backend hosting
Technical / hardware cost	Industrial GPUs, embedded vision processors, sensor arrays	YOLOv8n nano model + INT8 quantization; runs on ordinary CPUs and mobile browsers
Deployment cost	Conveyor belts, robotic arms, sorting stations (capital infrastructure)	Deployed on devices citizens already own (smartphones); no new hardware purchased
Maintenance cost	Physical machinery upkeep, specialised technician call-outs	Software-only maintenance: bug fixes, server upkeep, periodic model updates
Practicality / usability	Requires trained plant operators; not citizen-facing	Mobile-first PWA usable by non-technical citizens with minimal training

This comparison shows that WasteDetect achieves its low-cost positioning not by sacrificing core functionality, but by deliberately substituting expensive physical infrastructure with lightweight software components; a direct realisation of the brainstormed design direction established before any implementation work began.

4.5.3 CONTRIBUTION TO ENGINEERING AND TECHNOLOGY

WasteDetect demonstrates that an accurate, real time waste classification system can be built and deployed using only open source tools, consumer-grade mobile devices, and a model small enough to run without a GPU. This represents a practical engineering contribution: rather than proposing a marginal accuracy improvement over existing YOLOv8 research, this study contributes a complete, working, end-to-end system; covering data preparation, model optimisation, and a deployable PWA with offline support, explicitly tailored to the operational and financial constraints of a developing country deployment context such as Buea, Cameroon.

4.6 PARTIAL CONCLUSION

This chapter presented the implementation of the WasteDetect system and the results obtained from training, optimising, and evaluating the underlying YOLOv8n model, alongside the completed web application interfaces. The INT8-quantized model achieved a final test accuracy (mAP@50) of 0.598 at less than one-fifth the file size of the FP32 baseline, with comparable accuracy to other YOLOv8-based waste detection systems reported in the literature. The completed system was shown to satisfy the objectives set out in Chapter 1 and to achieve its low-cost design goal through deliberate software-based substitution of expensive physical infrastructure. Chapter 5 will conclude the dissertation with a summary of findings, the project's contribution to engineering and technology, recommendations, difficulties encountered, and directions for future work.

CHAPTER 5. CONCLUSION AND FURTHER WORKS

5.1 SUMMARY OF FINDINGS

This dissertation set out to design and implement a low-cost object detection system for waste sorting and recycling (WasteDetect), suitable for resource-constrained environments such as Buea, Cameroon. The work was motivated by two observations established in Chapter 1 and Chapter 2: waste management in Cameroon remains largely manual, with no institutional sorting mechanism in place at organisations such as HYSACAM; and existing AI-based waste sorting systems documented in the literature depend on expensive infrastructure; GPUs, robotic arms, and conveyor systems that makes them impractical for developing country deployment.

The field investigation conducted at HYSACAM, Buea, confirmed that no systematic waste sorting currently takes place, and identified plastic, metal, and organic as the categories of greatest operational interest, with paper and glass as desirable additions. These findings directly shaped the five-class waste categories used throughout the study and validated the low-cost design direction established during the brainstorming phase in Chapter 3.

A YOLOv8n object detection model was fine-tuned via transfer learning on a re-split and class-remapped Kaggle dataset of 10,464 images, with the training split expanded through five augmentation pipelines to 43,869 images to compensate for the limited environmental variability typically found in public waste datasets. Post-training INT8 quantization, applied through the OpenVINO toolkit, reduced the model's file size from 17.52 MB to 3.00 MB, an 82.9% reduction while keeping the drop in accuracy (mAP@50) to just 1.4%. On the held-out test split, the final INT8 model achieved an accuracy (mAP@50) of 0.598, a result that compares favourably with several YOLOv8-based waste detection studies reported in the literature, and sits only marginally below the result achieved by a larger, heavier YOLOv8s-based system.

Beyond the model itself, the study delivered a complete, working web-based system; WasteDetect, that captures a waste image via a mobile browser, classifies it in real time, and instructs the user where to place the item, closing the action loop that is often missing from purely informational classification tools. The system was built entirely from open source software, directly fulfilling the cost-conscious design direction set out from the project's earliest planning stages.

Taken together, these findings confirm the four research hypotheses stated in Chapter 1: a lightweight YOLOv8n model can achieve acceptable waste classification performance at low computational cost; image augmentation meaningfully improves model robustness; INT8 quantization substantially reduces model size and deployment cost without major accuracy loss; and a web-based, human-assisted system represents a more practical and cost effective solution for resource-constrained environments than a fully automated alternative.

5.2 CONTRIBUTION TO ENGINEERING AND TECHNOLOGY

Chapter 4 already established, in direct and precise terms, that this study's primary engineering contribution lies not in a marginal accuracy improvement over existing YOLOv8 research, but in the delivery of a complete, working, end-to-end system explicitly tailored to the financial and operational constraints of a developing country context. This section builds on that statement and unpacks, in more conversational terms, what that contribution actually means in practice and why it matters.

Most of the academic literature reviewed in Chapter 2 treats waste classification as a modelling problem: a new architecture, a new loss function, or a new dataset is proposed, and success is measured purely by whether accuracy improves on a benchmark. That is a valuable kind of research, but it rarely asks whether the resulting system could actually be afforded, hosted, and used by the people who most need it. This project deliberately asked that second question first. Every choice made in this study; the nano-sized YOLOv8n model instead of a larger variant, INT8 quantization instead of full-precision deployment, a browser-based PWA instead of a native app requiring an app store, and open source hosting on Vercel and Railway instead of proprietary cloud infrastructure was made specifically because it kept the system affordable and deployable in a context like Buea, where neither GPUs nor dedicated sorting hardware are realistic assumptions.

In doing so, the project demonstrates something that is easy to state but less often shown in practice: that a meaningful share of the value of AI-based waste sorting can be captured without the industrial infrastructure that the existing literature treats as a near requirement. WasteDetect achieves an accuracy (mAP@50) broadly comparable to systems built on heavier models and more expensive infrastructure, while running entirely on consumer-grade mobile devices and low-cost cloud hosting. This is the project's clearest technological contribution: evidence, grounded in a real field study and a real working deployment, that low-cost and reasonably accurate are not mutually exclusive goals in this domain.

There is also a contribution at the system's engineering level rather than the model level alone. The project did not stop at producing a trained model; it carried that model through the full path from a Kaggle notebook to a deployed, JWT-authenticated, offline-capable web application with its own database schema, admin dashboard, and bin placement guidance for the end user. This full stack realisation; spanning data engineering, model optimisation, backend API design, and frontend PWA development is itself a demonstration of applied computer engineering practice, and represents the kind of integration work that a purely academic model-accuracy paper would not typically undertake.

Finally, the project contributes a grounded, field-validated waste category and operational understanding for the Buea context specifically something that did not previously exist in a form usable for AI model training. The questionnaire and interview conducted at HYSACAM produced first hand evidence of local waste collection methods, operational challenges, and category priorities that can inform not only this project but future research or systems targeting the same region.

5.3 RECOMMENDATIONS

Based on the experience of designing, building, and evaluating WasteDetect, the following recommendations are made for organisations, researchers, and developers considering similar low-cost AI-based waste sorting initiatives:

- **Invest in locally collected training data.** Although the publicly available Kaggle dataset used in this study was sufficient to demonstrate feasibility, a model trained on images of waste as it actually appears in Buea, mixed, weathered, and photographed under local lighting conditions would likely generalise better than one trained primarily on cleaner, internationally sourced images.
- **Engage waste management organisations early and formally.** The field study at HYSACAM produced valuable findings, but the administrative process of gaining access took several weeks. Future researchers and organisations should establish a formal data sharing or research partnership with HYSACAM (or similar bodies) ahead of time, rather than negotiating access on a per-project basis.
- **Pilot the system with a small group of real users before wider rollout.** WasteDetect has been evaluated on held-out test data and through manual testing of the interface, but it has not yet been tested with real citizens sorting real waste over an extended period. A small scale pilot would surface usability issues and confidence-threshold calibration needs that are difficult to anticipate from offline evaluation alone.
- **Treat the low-confidence mismatch log as an active improvement resource, not just an audit trail.** The system already logs low-confidence predictions and user corrections. If WasteDetect is adopted operationally, this log should be periodically reviewed and used to guide targeted retraining or data collection, rather than left unreviewed.
- **Consider a phased rollout of additional languages and categories.** The current MVP is scoped to English and five waste categories. Any expansion to French-language support or to additional categories such as e-waste (electronic-waste such as batteries) should be done incrementally and validated against real user feedback rather than added speculatively.

5.4 DIFFICULTIES ENCOUNTERED

Several practical difficulties were encountered over the course of this project, each of which shaped a design or scoping decision in this dissertation.

- **Acquiring data that reflects the local context.** No publicly available dataset captures waste as it actually appears in Cameroon; typically mixed rather than cleanly separated, and photographed under local environmental conditions rather than in a studio setting. The dataset ultimately used in this study, sourced from Kaggle, was the closest practical substitute available within the project's time and resource constraints, but it does not fully reflect the visual conditions a WasteDetect user in Buea would encounter. This limitation was partially addressed through the five pipeline image augmentation strategy described in Chapter 3, which artificially introduced lighting and noise variation, but augmentation cannot fully substitute for genuinely local training data.
- **The field study process at HYSACAM was unexpectedly demanding.** As detailed in Chapter 3, what was anticipated to be a single visit instead required three separate visits requiring nearly three weeks: an initial visit that was turned away for lack of authorisation, a return trip to collect the university authorisation letter, a second visit at HYSACAM that additionally required an on-the-spot handwritten letter and a typed questionnaire (drafted under time pressure in three to four hours), and a final visit to collect the completed questionnaire after a further two hour wait. This process, while ultimately successful, was administratively and logistically stressful, and consumed time that could be allocated to other phases of the project.
- **Balancing model accuracy against the low-cost design constraint.** Every stage of model development involved a tradeoff between detection accuracy and deployability. A larger model or a longer training run may have produced a marginally higher accuracy (mAP@50), but would have worked against the project's central low-cost objective. Deciding where to draw this line for example, choosing YOLOv8n over YOLOv8s, or

accepting a 1.4-percentage-point accuracy drop in exchange for an 82.9% reduction in model size required repeated judgement calls rather than a single clear-cut answer.

- **Working within a single developer timeline.** This project covered field research, dataset engineering, model training and optimisation, and full stack web development, all carried out by one person within an academic year timeline. Each of these areas alone typically warrants specialised attention, and sequencing them realistically without allowing any one phase to consume time needed by the others was a recurring challenge throughout the project.

5.5 FURTHER WORKS

The current version of WasteDetect satisfies the Must-Have features defined in Chapter 3 and successfully demonstrates the feasibility of a low-cost, human-assisted waste classification system. However, several aspects of the original requirements specification remain open for future development, and additional opportunities for improvement have been identified over the course of this work.

5.5.1 COMPLETE THE ADMIN INTERFACES AND FUNCTIONALITIES

The most immediate piece of further work is the completion of the admin pages and functionality defined in the requirements specification and Must-Have features of Chapter 3, but not yet implemented in the current version of the system. Specifically, future work should deliver:

- An Admin Dashboard interface displaying the total number of items sorted for the current day and month.
- A confidence breakdown view showing the percentage of correctly versus incorrectly sorted items.
- Chart.js-based visualisations; bar, pie/doughnut, and line charts presenting sorting statistics to the admin.
- A mismatch and confidence log viewer, allowing the admin to review the auto-generated explanations attached to low-confidence classifications.

- Route level access restriction ensuring the admin dashboard and its underlying endpoints remain accessible only to authenticated admin users.

Completing these admin interfaces would fully satisfy the system requirements scoped in Chapter 3 and would give HYSACAM officials direct, practical visibility into sorting activity a capability that was one of the original motivations for the project, identified during the field study at HYSACAM.

5.5.2 OTHER DIRECTIONS FOR FUTURE WORK

- **Locally collected training data:** Future iterations should incorporate images captured directly in Buea or similar Cameroonian urban contexts, reducing reliance on internationally sourced datasets and improving real world classification accuracy.

- **In-app retraining pipeline:** Currently, the YOLOv8n INT8 model has no in-app retraining. A future version could use the accumulated low-confidence and corrected classification logs as a basis for periodic model retraining, closing the feedback loop between deployment and model improvement.

- **Multi language support:** Extending the user interface to French, in addition to the English-only MVP scope, would improve accessibility across Cameroon's two official languages.

- **Pilot deployment and field evaluation:** A structured pilot with real users in Buea, potentially in partnership with HYSACAM, would validate the system's usability and classification performance under genuine field conditions rather than offline test-set evaluation alone.

- **Expanded waste category coverage:** Future versions could extend beyond the five current categories to include categories such as e-waste or hazardous waste, subject to further field validation of demand for these categories.

Pursued together, these directions would extend WasteDetect from a working academic prototype into a system genuinely ready for sustained real world use without departing from the low-cost design principles that have guided this project from its earliest brainstorming stage through to its final implementation.

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APPENDICES

I. AUTHORIZATION LETTER OBTAINED FROM SCHOOL

UNIVERSITY OF BUEA
P.O. Box 63,
Buea, South West Region
CAMEROON
Tel: (237) 3332 21 34/3332 26 90/3332 27 06/3332 28 13
Fax: (237) 3332 22 72

REPUBLIC OF CAMEROON
PEACE-WORK-FATHERLAND

FACULTY OF ENGINEERING AND TECHNOLOGY

Dean: **Agbor Dieudonne Agbor, PhD**
Vice-Dean/PAA: **Nde D. Nguti, PhD**
Vice-Dean/SAR: **Ngwashi Divine Khan, PhD**
Vice-Dean/RC: **Tsafack Pierre, PhD**
Faculty Officer: **Nyeka-Sanga nee Emeli Akanang Eunice**

No. 2026/ /UB/FET/HOD/CE

Date: 05 MAY 2026

FROM: HOD Computer Engineering

TO: Whom it may concern

Object: Authorization to carry out information collection for a project course at **HYSACAM Headquarter of Buea**

Dear Sir/Madam,

This letter serves to formally authorize **ATANKEU TCHAKOUTE ANGE (FE22A158)**, Level **500** students in the **Department of Computer Engineering, Faculty of Engineering and Technology, University of Buea**, to carry out data and information collection at your institution for his final year project.

Their research is titled: **"Low Cost Object Detection for Waste Sorting and Recycling"**.

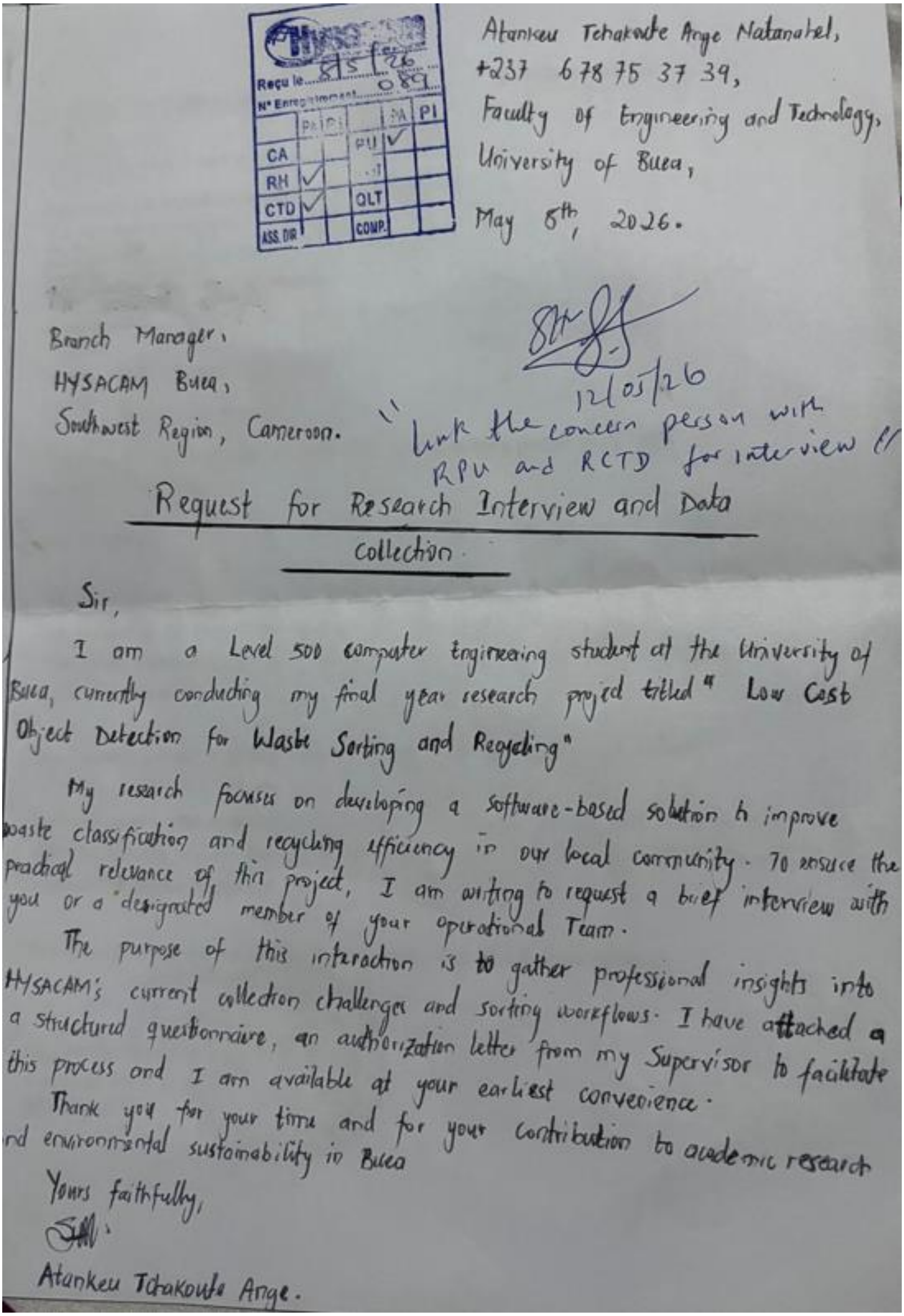
We kindly request your support in facilitating their work by him to collaborate with the relevant staff. Additionally, we ask that he can be granted access to the necessary information to successfully complete his research.

We highly appreciate your cooperation and thank you in advance for your assistance.

REPUBLIC OF CAMEROON
FACULTY OF ENGINEERING AND TECHNOLOGY
Department of Computer Engineering
H.O.D
UNIVERSITY OF BUEA

Elie FUTE TAGNE, PhD
Associate Professor
HOD/CE-FET

II. APPLICATION LETTER



Reçu le 8/5/26
N° Enregistrement 089

Atankeu Tchakoute Ange Natanahel,
+237 678 75 37 39,
Faculty of Engineering and Technology,
University of Buea,
May 5th, 2026.

Branch Manager,
HYSACAM Buea,
Southwest Region, Cameroon.

12/05/26
"Link the concern person with RPU and RCTD for interview"

Request for Research Interview and Data collection

Sir,

I am a Level 500 computer engineering student at the University of Buea, currently conducting my final year research project titled " Low Cost Object Detection for Waste Sorting and Recycling"

My research focuses on developing a software-based solution to improve waste classification and recycling efficiency in our local community. To ensure the practical relevance of this project, I am writing to request a brief interview with you or a designated member of your operational Team.

The purpose of this interaction is to gather professional insights into HYSACAM's current collection challenges and sorting workflows. I have attached a structured questionnaire, an authorization letter from my Supervisor to facilitate this process and I am available at your earliest convenience.

Thank you for your time and for your contribution to academic research and environmental sustainability in Buea

Yours faithfully,

Atankeu Tchakoute Ange.

III. RESEARCH QUESTIONNAIRE

Research Questionnaire: Waste Management and Recycling Optimization

Introduction: This interview is conducted to gather professional insights into current waste collection and sorting workflows in Cameroon. The data collected will support the development of an academic project at the University of Buea aimed at implementing a low-cost, efficient waste-sorting solution to enhance local recycling efforts.

Section 1: General Collection Procedures

1. How do you currently manage the collection of waste within the municipality? (Select all that apply)
   - [✓] Door-to-door collection
   - [✓] Community skip/container collection
   - [✓] Scheduled curbside pickup
   - [✓] Informal/Private collectors
   - [ ] Other: ____________________

2. Have you or your team ever experienced specific issues or bottlenecks during the waste collection process?
   - [✓] Yes
   - [ ] No

3. If yes, could you please describe the nature of these challenges? (Select all that apply)
   - [ ] Fluctuating or limited power supply
   - [✓] Limited financial/budgetary resources
   - [✓] Poor road infrastructure/accessibility
   - [✓] Frequent vehicle breakdowns
   - [✓] Lack of public cooperation/illegal dumping
   - [ ] Other: ____________________

Section 2: Waste Sorting and Recycling Workflow

4. Does your organization currently sort waste materials specifically for recycling purposes?
   - [ ] **Yes** (Please proceed to Questions 5–9)
   - [✓] **No** (Please skip to Questions 10–11)

IF YES (Current Sorting Practices):

5. At what stage is the sorting performed?
   - [ ] Before collection (at the source)
   - [ ] After collection (at a sorting/transfer center)
6. Into which specific categories is the waste currently sorted? (Select all that apply)
   - [ ] Plastic
   - [ ] Paper / Cardboard
   - [ ] Metal / Aluminum
   - [ ] Organic / Food Waste
   - [ ] Glass
   - [ ] Other: ____________________
7. How frequently is the sorting process conducted?
   - [ ] Daily
   - [ ] Weekly
   - [ ] Monthly
   - [ ] On-demand
8. What are the primary technical or operational difficulties faced during the sorting process? (Select all that apply)
   - [ ] High contamination (waste mixed with mud/dirt)
   - [ ] Lack of automated sorting technology
   - [ ] Shortage of manual labor
   - [ ] Insufficient space for sorting
   - [ ] Other: ____________________
9. What is the final destination or use for the materials once they have been sorted?
   - [ ] Sold to local recycling plants
   - [ ] Exported for processing
   - [ ] Used for composting
   - [ ] Other: ____________________

IF NO (Prospective Sorting Interests):

10. Would your organization be interested in implementing a system to sort collected waste?
    - [ ] Very Interested
    - [✓] Somewhat Interested

- [ ] Neutral
- [ ] Not Interested

11. Which material categories do you believe are most critical to sort to facilitate effective recycling in this region (Select top 3)?

- [✓] Plastic
- [ ] Paper
- [✓] Metal
- [ ] Glass
- [✓] Organic Waste
- [ ] Other: ____________

Section 3: Technological Integration and Future Outlook

12. Would a technological solution that enables efficient sorting before collection and provides real-time data on waste volumes across different quarters be advantageous for your team?

- [ ] Highly Advantageous
- [✓] Somewhat Advantageous
- [ ] Neutral
- [ ] Not Advantageous

13. In your professional opinion, how would such a solution improve overall operations in terms of time management, cost-effectiveness, and convenience? (Select all that apply)

- [✓] Reduction in manual sorting time
- [✓] Better allocation of collection trucks based on volume data
- [✓] Improved convenience for the collection team
- [✓] Cost-effectiveness due to reduced waste volume in landfills
- [ ] Other: ____________________

Thank you for your time and professional contribution to this research.